Firms in Competitive Markets

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Slide 2

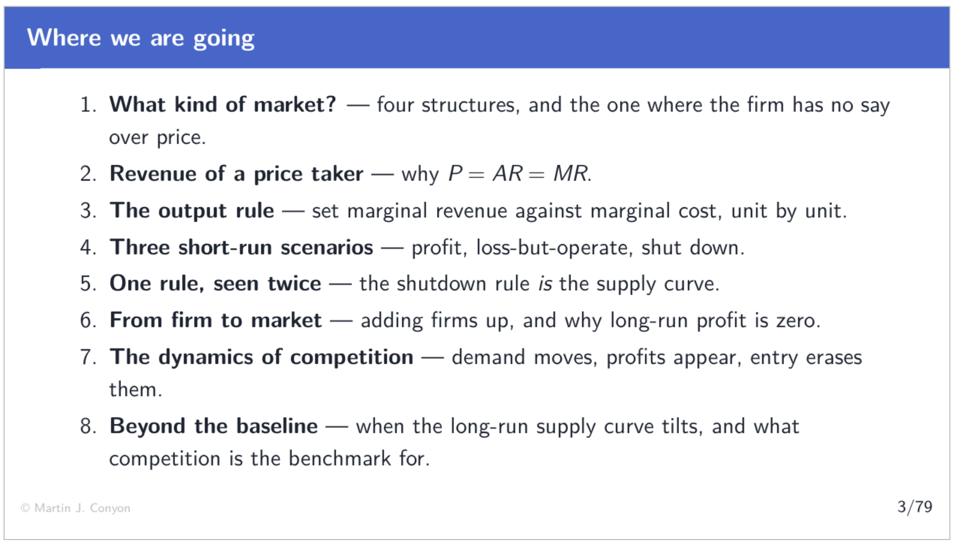
Working through it

Why this chapter. Mankiw's chapter 15 covers the same sequence: what makes a market competitive, why a price taker's marginal revenue equals the price, the profit-maximizing output rule, the shutdown and exit decisions, the firm's supply curve, and the long-run zero-profit outcome driven by entry and exit.

What the slides add. Three things. Marginal revenue and marginal cost are built as *columns of differences* rather than derivatives, so the output rule is arrived at unit by unit. The shutdown rule and the supply curve are taught as *one* result rather than two, which is where most confusion in this topic comes from. And the long-run adjustment is computed to the individual firm — including the whole-firm check that shows exactly why entry stops where it does, rather than at a convenient round number.

How to use it. Read the chapter before for vocabulary and after for consolidation. The chapter states that entry drives profit to zero; the slides show you the arithmetic of the last firm in.

Where we are going



Slide 3

Working through it

The structure. Eight parts in two clear halves.

Parts 1–5: one firm alone. What kind of market it is in (1), what its revenue looks like (2), the rule that fixes its output (3), what happens at three very different prices (4), and the discovery that those answers, collected, *are* its supply curve (5).

Parts 6–8: the market. Adding firms up and the long-run zero-profit condition (6); then the centerpiece, Part 7, where demand is shocked in both directions and price, profit and the number of firms are all computed at every stage; then two honest qualifications and the benchmark claim (8).

The single thread. Every answer comes from comparing the market price with the firm's own cost curves. Nothing else is ever needed.

1. Part 1: What Kind of Market?

Four ways to organize a market

Four ways to organize a market

Structure	Sellers	Product	Firm's power over price
Perfect competition	many	identical	none
Monopolistic competition	many	differentiated	a little
Oligopoly	few	either	considerable
Monopoly	one	unique	the most

Everything here lives on the **top row**. The other three come later.

The fewer the sellers, the more each one can say about the price.

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Slide 5

Working through it

Why classify before analyzing. How a firm behaves depends on how much power it has over its own price, and that depends on the structure of the market around it. Studying the firm without first fixing the structure produces answers that are true in one setting and false in another.

The two questions that generate the table. How many sellers are there, and do they sell the same thing?

The four structures.

- **Perfect competition**: many sellers, identical product, no price power whatever.
- **Monopolistic competition**: many sellers, differentiated product, so a little price power — a firm can raise its price slightly without losing everyone.

- **Oligopoly**: few sellers, so each is large enough that its own decisions move the market.
- **Monopoly**: one seller, the most power available.

The pattern. The fewer the sellers and the more distinctive the product, the more each firm can say about the price. This material takes the top row, where the firm can say nothing at all — which makes it the cleanest case and, as Part 8 argues, the benchmark for all the others.

The three conditions

Slide 6

Working through it

The definition. A competitive market has (1) many buyers and many sellers, (2) identical goods across sellers, and (3) free entry and exit.

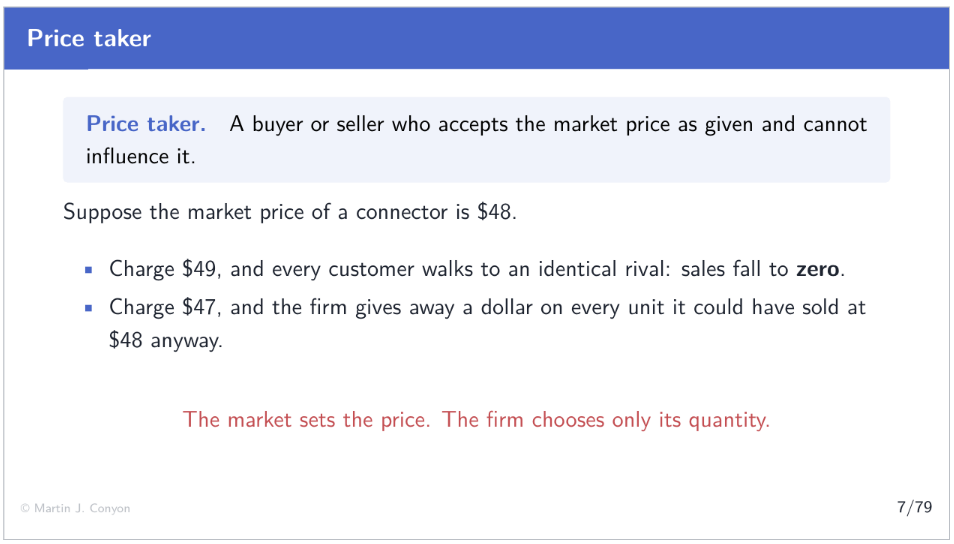
What each condition is doing, because none is decorative.

- **Many sellers** ensures no firm is a large enough share of total output to move the price by changing its own production. This is what makes the firm a price *taker*.
- **Identical goods** ensures buyers switch supplier for a penny. Without it, a firm could raise its price a little and keep some customers — which is exactly monopolistic competition.

- **Free entry and exit** ensures the *number of firms* can change. Conditions 1 and 2 govern the short run; this one governs the long run, and it is the engine of Parts 6 and 7.

The running example. A maker of standardized electronic connectors — a part where one supplier's output is genuinely indistinguishable from another's, so all three conditions are plausible at once.

Price taker



Slide 7

Working through it

The definition. A **price taker** accepts the market price as given and cannot influence it.

Why the three conditions force it. Suppose the market price is $48.

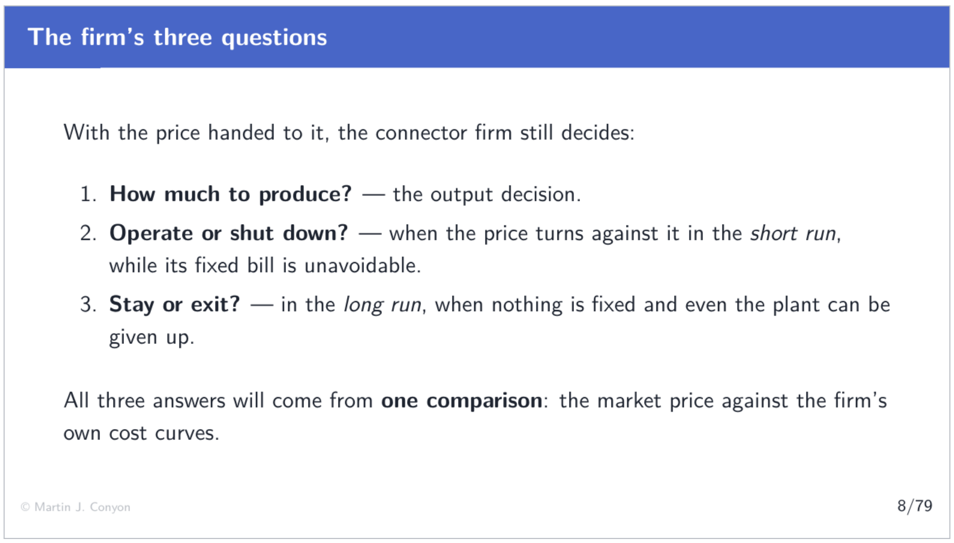
- **Charge $49.** The product is identical and there are many sellers, so every buyer moves to a rival at $48. Sales fall to **zero** — not "fall somewhat", but to nothing.

- **Charge $47.** The firm is small enough to sell its entire output at $48, so cutting the price wins no extra sales it could not already make. It simply hands a dollar to each buyer.

So the firm's decision problem is smaller than it looks. Price is *data*, handed down by the market. The only variable the firm controls is quantity.

The trap this sets up, and Active Learning 2 walks into it deliberately. Advice like "undercut to win customers" is sound only for a firm whose sales are limited by its price. A price taker's are not — it can already sell everything it makes — so price cutting is a pure transfer to buyers.

The firm's three questions



Slide 8

Working through it

What remains once the price lever is gone. Three genuine decisions, and the parts ahead answer them in order.

1. How much to produce? The output decision, taken at whatever price prevails. Part 3.

2. Operate or shut down? A *short-run* question, asked when the price turns against the firm while its fixed bill remains unavoidable. Part 4.

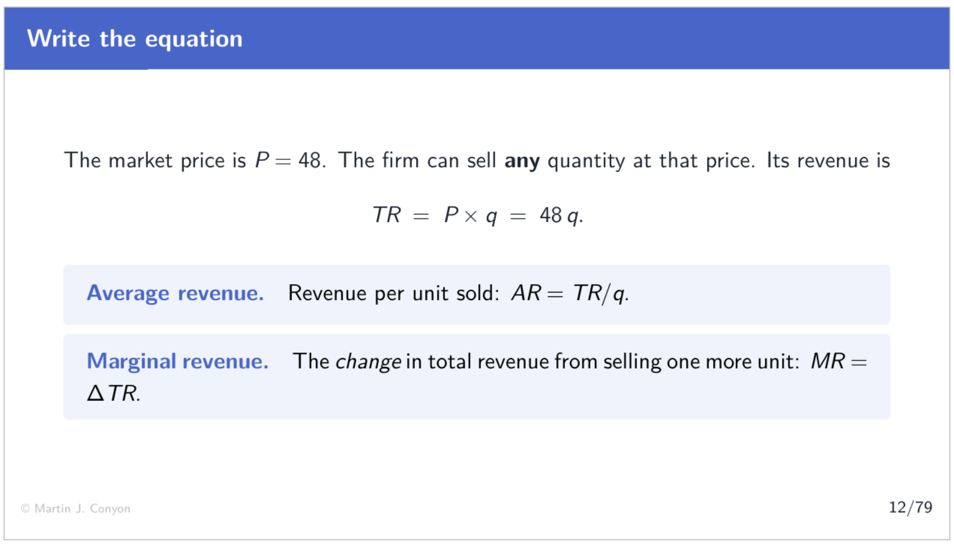
3. Stay or exit? A *long-run* question, asked when nothing is fixed any more and even the plant can be given up. Part 5.

Why the second and third are different questions. Shutting down means producing nothing while still paying the rent. Exiting means the rent stops too. They compare the price against different costs, and getting them confused is the single most common error in this topic.

The unifying claim. All three answers come from one comparison — the market price held against the firm's own cost curves. No new machinery is required beyond the cost measures already built.

2. Part 2: Revenue of a Price Taker

Write the equation



Slide 12

Working through it

The revenue side, which is the easy half of profit. The market price is \$48 and the firm can sell *any* quantity at it, so

$$TR = P \times q = 48q$$

Two per-unit measures, mirroring the cost measures exactly.

- **Average revenue**: $AR = TR/q$ — revenue per unit sold.

- **Marginal revenue**: $MR = \Delta TR$ — the change in total revenue from selling one more unit.

Why marginal revenue is the one that matters. The firm's decision is about the *next* unit, so it needs to know what the next unit brings in. Average revenue describes units already sold, exactly as average cost describes units already produced.

What to watch for in the table. For this firm both measures will turn out to equal the price, in every row. That coincidence is not general — it is a property of price taking, and the frame after the table separates the part that is trivial from the part that is not.

Plug the numbers in

Plug the numbers in

q	$TR = 48 \times q$	$AR = TR/q$	$MR = \Delta TR$
0	$48 \times 0 = 0$	—	—
1	$48 \times 1 = 48$	$48/1 = 48$	48
2	$48 \times 2 = 96$	$96/2 = 48$	48
3	$48 \times 3 = 144$	$144/3 = 48$	48
4	$48 \times 4 = 192$	$192/4 = 48$	48
5	$48 \times 5 = 240$	$240/5 = 48$	48
6	$48 \times 6 = 288$	$288/6 = 48$	48
7	$48 \times 7 = 336$	$336/7 = 48$	48
8	$48 \times 8 = 384$	$384/8 = 48$	48

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Slide 13

Working through it

Building the table. $TR = 48q$, so at $q = 3$, $TR = 144$; at $q = 7$, $TR = 336$.

Average revenue. $144/3 = 48$; $336/7 = 48$. Constant at 48, which is unsurprising: every unit sells at \$48, so revenue per unit is \$48.

Marginal revenue — the column to study. Differences of the TR column:

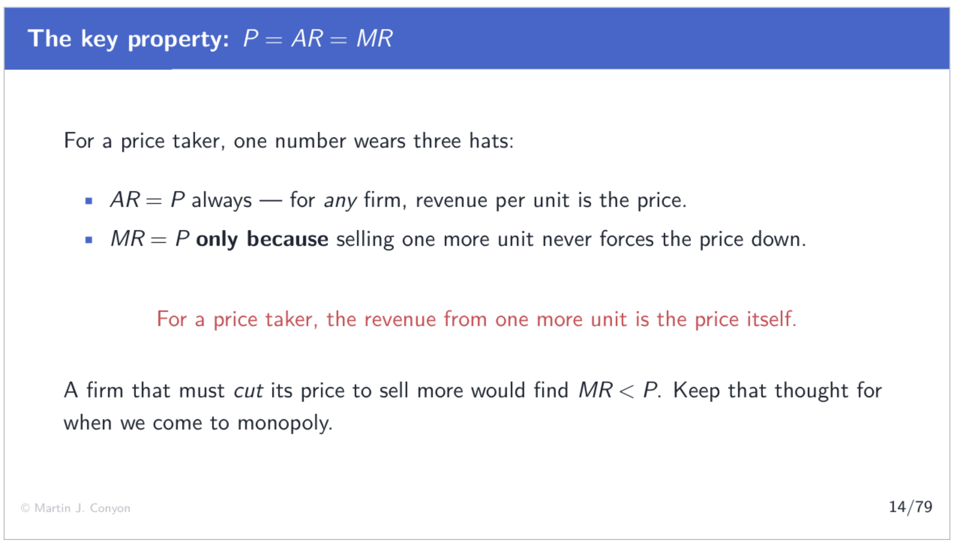
$$48 - 0 = 48 \qquad 96 - 48 = 48 \qquad 336 - 288 = 48$$

Every jump is \$48. The third unit adds \$48; so does the eighth.

Why the marginal column is constant, stated carefully. Selling one more unit never requires cutting the price on the units already being sold. So the extra revenue is simply the price of the extra unit, with no offsetting loss anywhere.

Why that is worth stating. It sounds too obvious to mention, and it is false for most firms. A seller that must lower its price to move more units gives up revenue on *all* its units, and its marginal revenue falls short of the price accordingly. The next frame isolates exactly which half of $P = AR = MR$ is special.

The key property: $P = AR = MR$



Slide 14

Working through it

One number, three hats — but the three are not equally interesting.

$AR = P$ **always.** For any firm at all, revenue per unit is the price per unit. This is arithmetic, not economics: $TR/q = Pq/q = P$. It carries no information.

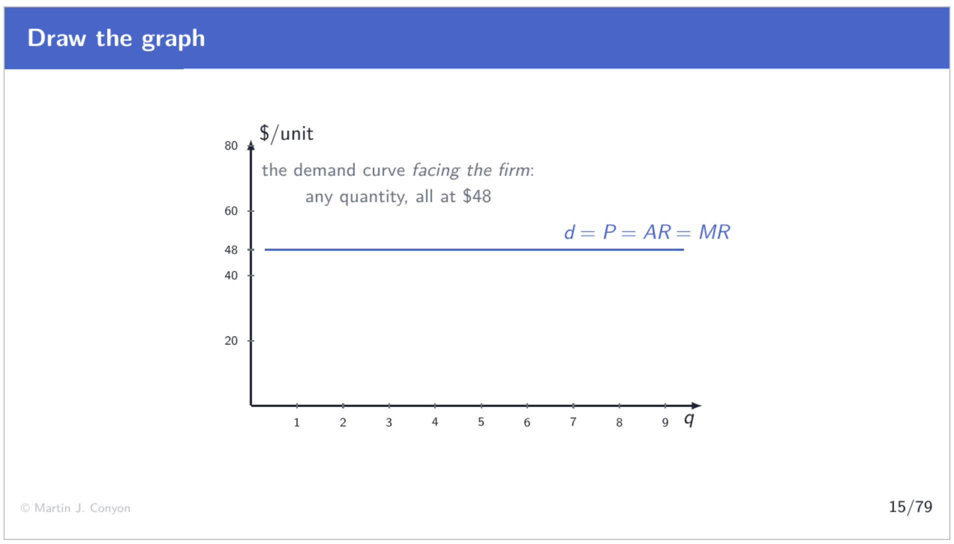
$MR = P$ **only for a price taker.** This is the substantive claim, and it holds precisely because selling one more unit does not force the price down.

The contrast that makes it concrete. Consider a firm facing the whole market demand curve. To sell one extra unit it must cut the price — and the cut applies to every unit it was already selling. So

$$MR = \underbrace{P}_{\text{from the extra unit}} - \underbrace{(\text{price cut}) \times (\text{units already sold})}_{\text{lost on the rest}} < P$$

Why this matters now rather than later. The output rule ahead is $MR = MC$, which is universal. Only for a price taker does it simplify to $P = MC$. When we come to monopoly, the whole difference between the two market structures traces back to this one wedge.

Draw the graph



Slide 15

Working through it

What is drawn. The demand curve *facing the individual firm*: a horizontal line at $48.

How to read a horizontal demand curve. At $48 the firm can sell as much as it likes — the market absorbs its entire output without the price moving. At a fraction above $48 it sells nothing. So the line is horizontal at the market price and the firm's whole feasible set lies along it.

Why one line carries three labels. Because the price is the same for every unit, the height of this line is simultaneously the price, the average revenue and the marginal revenue.

$d = P = AR = MR$.

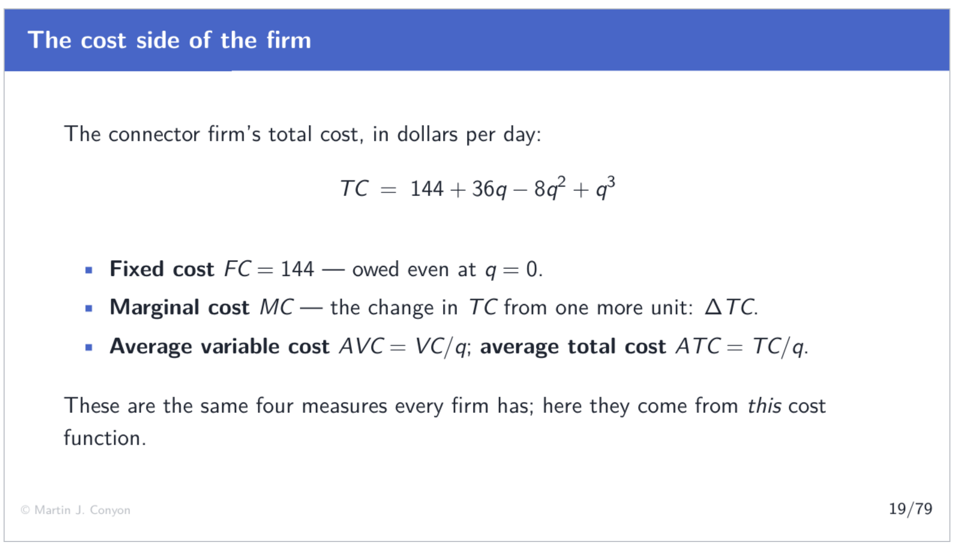
The distinction to keep firmly in mind. There are *two* demand curves in this material, and confusing them is a standard error.

- The **market** demand curve slopes down, as it always has. Buyers collectively take more only at a lower price.
- The demand curve facing **one firm** is flat, because that firm is a negligible part of the market.

Both are true at once. The flat line is what a downward-sloping market curve looks like from the vantage of a firm too small to move along it.

3. Part 3: The Output Rule

The cost side of the firm



Slide 19

Working through it

The other half of profit. The firm's technology is summarized in one function:

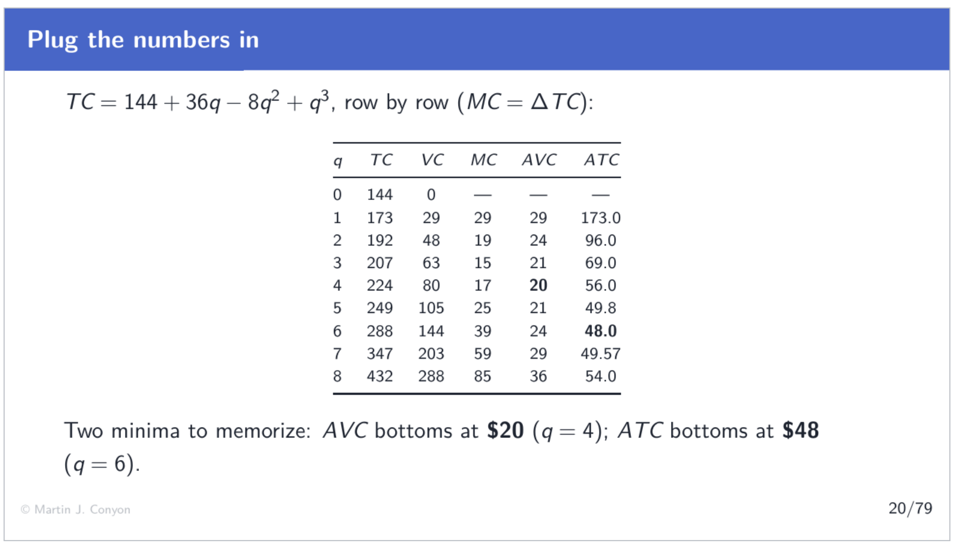
$$TC = 144 + 36q - 8q^2 + q^3$$

Reading it. The **144** is the fixed bill — rent and machines, owed even at $q = 0$. The rest is variable, and its cubic shape produces marginal cost that falls at first and then climbs, which is the standard short-run pattern.

The measures taken from it. Marginal cost as ΔTC; average variable cost as VC/q; average total cost as TC/q. These are the same four measures every firm has — nothing new is being introduced.

What is new. Not the cost side, which is already familiar, but what is about to be done with it: held up against the revenue side, unit by unit, to produce an output decision. Everything from here to the end of the material is that comparison, run at different prices and then at the level of a whole industry.

Plug the numbers in



Plug the numbers in

$TC = 144 + 36q - 8q^2 + q^3$, row by row ($MC = \Delta TC$):

q	TC	VC	MC	AVC	ATC
0	144	0	—	—	—
1	173	29	29	29	173.0
2	192	48	19	24	96.0
3	207	63	15	21	69.0
4	224	80	17	**20**	56.0
5	249	105	25	21	49.8
6	288	144	39	24	**48.0**
7	347	203	59	29	49.57
8	432	288	85	36	54.0

Two minima to memorize: AVC bottoms at **$20** ($q = 4$); ATC bottoms at **$48** ($q = 6$).

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Slide 20

Working through it

Auditing one row, $q = 6$. $TC = 144 + 216 - 288 + 216 = 288$. So $VC = 288 - 144 = 144$, $AVC = 144/6 = 24$, $ATC = 288/6 = 48$, and $MC = 288 - 249 = 39$.

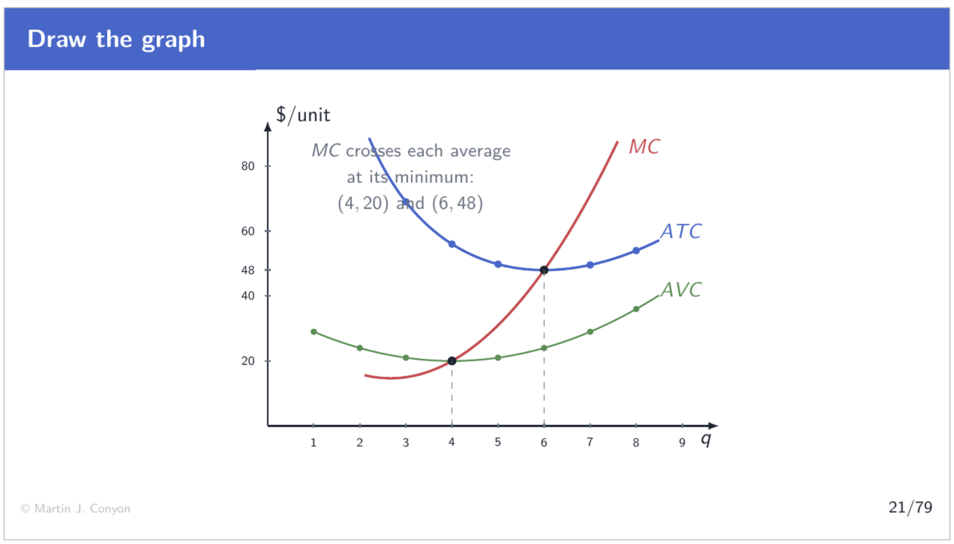
Reading the marginal cost column. 29, 19, 15, 17, 25, 39, 59, 85. It falls to $15 at the third unit, then climbs steeply as diminishing returns take hold. That U-shape is what makes the output decision interesting.

The two numbers to memorize, because every decision ahead uses one of them.

- **min** $AVC = \$20$, at $q = 4$. The *shutdown* trip-wire.
- **min** $ATC = \$48$, at $q = 6$. The *exit* trip-wire, and the long-run price of the whole industry.

Why these two and no others. Every question the firm faces reduces to where the market price sits relative to one of them. Above 48: profit, and entrants come. Between 20 and 48: operate at a loss, and plan to leave. Below 20: shut down. The entire decision map is two numbers.

Draw the graph



Slide 21

Working through it

The same logic in a picture. The flat line is what every unit brings in; the rising curve is what each unit costs.

Reading the two arrows.

- **Left of** $q = 6$: the price line lies above marginal cost, so each additional unit adds more than it costs. The green arrow points outward — expand.

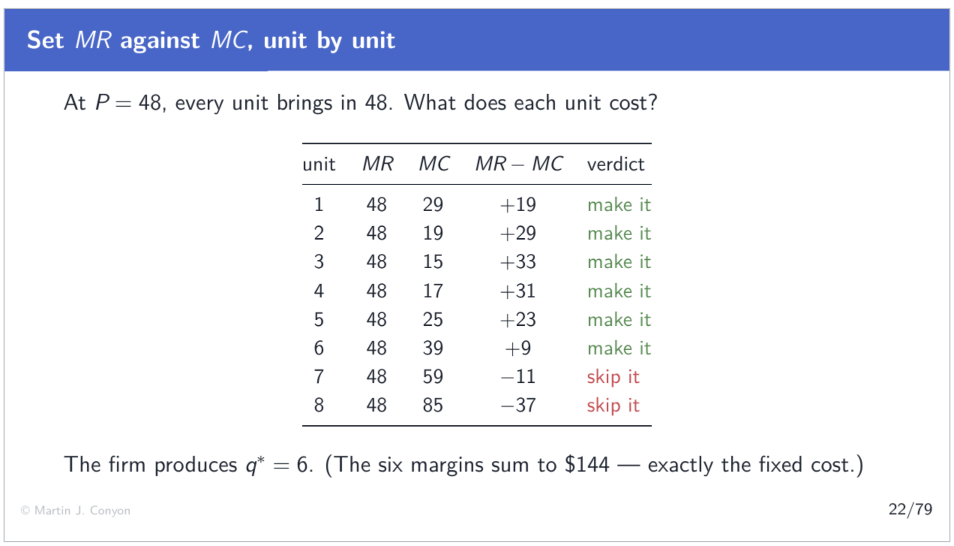
- **Right of** $q = 6$: marginal cost lies above the price, so the last unit lost money. The red arrow points inward — cut back.

The two pressures squeeze output onto the crossing point.

Why the crossing is a maximum rather than a minimum. Because marginal cost is *rising* where it meets the price. If the price line crossed a falling marginal cost curve, the same reasoning would push output away from the crossing rather than toward it.

The image to hold, because Part 5 depends on it. When the market price changes, the horizontal line slides up or down and the crossing point slides along the marginal cost curve. A menu of prices therefore traces out a menu of quantities — and that is a supply curve, though nobody has said so yet.

Set MR against MC, unit by unit



Set MR against MC, unit by unit

At $P = 48$, every unit brings in 48. What does each unit cost?

unit	MR	MC	$MR - MC$	verdict
1	48	29	+19	make it
2	48	19	+29	make it
3	48	15	+33	make it
4	48	17	+31	make it
5	48	25	+23	make it
6	48	39	+9	make it
7	48	59	−11	skip it
8	48	85	−37	skip it

The firm produces $q^* = 6$. (The six margins sum to \$144 — exactly the fixed cost.)

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Slide 22

Working through it

The comparison, unit by unit. At $P = 48$ every unit brings in \$48. What does each unit cost?

- Unit 1: $48 - 29 = +19$. Make it.
- Unit 3: $48 - 15 = +33$. The best unit of the day.

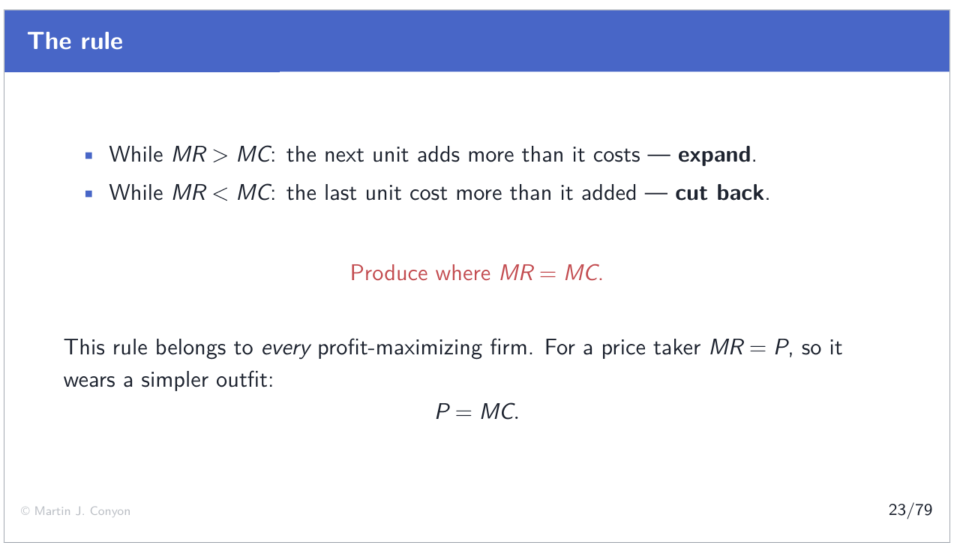
- Unit 6: $48 - 39 = +9$. Still worth making, though only just.
- Unit 7: $48 - 59 = -11$. Skip it.

So $q^* = 6$. The firm produces every unit whose marginal cost the price covers, and stops at the first one it does not.

Why the answer cannot be improved. Producing a seventh unit would add \$48 of revenue and \$59 of cost — a loss of \$11. Producing only five would forgo the \$9 the sixth unit contributes. Six is the peak, and the table shows it without any calculus.

The arithmetic curiosity at the foot of the frame, which is not a curiosity. The six positive margins sum to $19 + 29 + 33 + 31 + 23 + 9 = \$144$ — exactly the fixed cost. So at \$48 the operating surplus covers the rent to the penny and no more, which means profit is exactly zero. That is not a coincidence: \$48 is the minimum of ATC, and the next frames build the whole second half of the material on that fact.

The rule



Slide 23

Working through it

The rule, in two halves.

- While $MR > MC$: the next unit adds more than it costs, so **expand**.

- While $MR < MC$: the last unit cost more than it added, so **cut back**.

The profit peak is therefore where $MR = MC$.

Why this is universal. Nothing in the argument used price taking, market structure, or the shape of the cost curves. Any firm maximizing profit obeys $MR = MC$ — monopolists included.

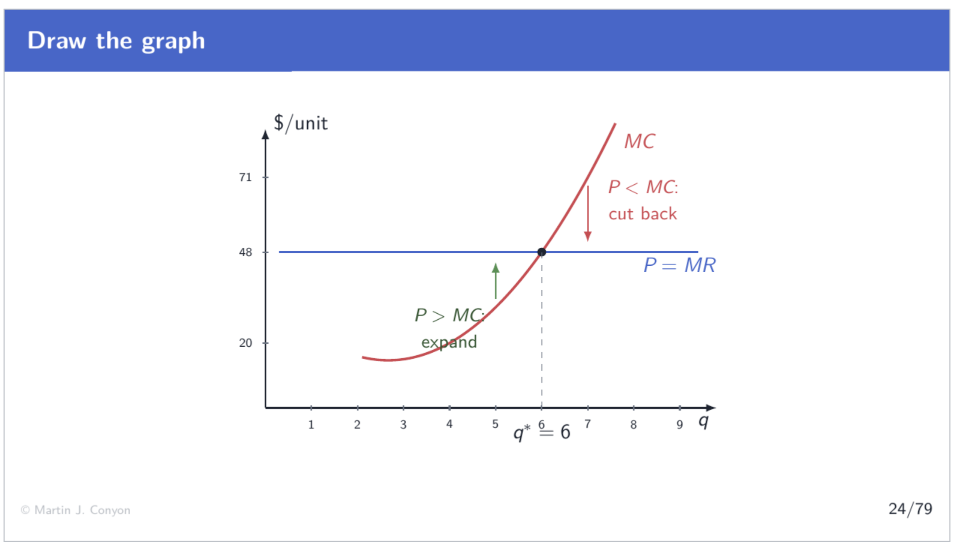
The special costume for a price taker. Because $MR = P$ here, the rule reads

$$P = MC$$

which is why the firm can simply run a finger up its own marginal cost column and stop where the market price meets it.

The warning implicit in the distinction. $P = MC$ is a *consequence* of price taking, not the rule itself. Applied to a firm that must cut its price to sell more, it gives the wrong answer — because for that firm MR and P are different numbers.

Draw the graph



Slide 24

Working through it

The same logic in a picture. The flat line is what every unit brings in; the rising curve is what each unit costs.

Reading the two arrows.

- **Left of** $q = 6$: the price line lies above marginal cost, so each additional unit adds more than it costs. The green arrow points outward — expand.
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The two pressures squeeze output onto the crossing point.

Why the crossing is a maximum rather than a minimum. Because marginal cost is *rising* where it meets the price. If the price line crossed a falling marginal cost curve, the same reasoning would push output away from the crossing rather than toward it.

The image to hold, because Part 5 depends on it. When the market price changes, the horizontal line slides up or down and the crossing point slides along the marginal cost curve. A menu of prices therefore traces out a menu of quantities — and that is a supply curve, though nobody has said so yet.

Check it in totals

Check it in totals

The marginal route said $q^* = 6$. Verify with total profit, $\pi = 48q - TC$:

q	$TR = 48q$	TC	π
4	192	224	−32
5	240	249	−9
6	288	288	**0**
7	336	347	−11
8	384	432	−48

Both routes agree: the peak is at $q = 6$. And the peak is **exactly zero** — because 48 is the minimum of ATC. *That number will run the whole second half of the lecture.*

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Slide 25

Working through it

Why check. The marginal route gave $q^* = 6$. Totals are an independent computation, so agreement is evidence and disagreement would be an alarm.

The totals. $\pi = 48q - TC$:

$$q = 5: \; 240 - 249 = -9 \qquad q = 6: \; 288 - 288 = \mathbf{0} \qquad q = 7: \; 336 - 347 = -11$$

The peak is at $q = 6$, confirming the margins.

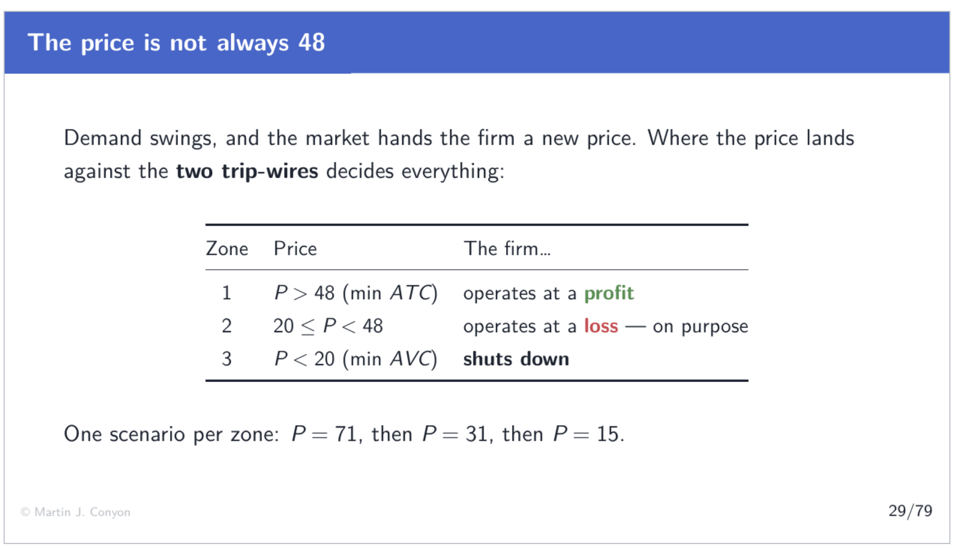
The value of the peak, which is the important part. Exactly *zero*. Not a loss and not a gain.

Why zero, and why it is not an accident. Profit per unit is $P - ATC$, and \$48 happens to be the *minimum* of ATC. So at its very best output the firm exactly covers every cost including the owner's normal return, and has nothing left over. At any other output ATC is higher than 48, so the firm would make a loss.

Where this is going. A price equal to min ATC is the only price at which a competitive industry can be at rest — above it, entrants come; below it, firms leave. That single number, 48, runs the entire second half.

4. Part 4: Three Short-Run Scenarios

The price is not always 48



Slide 29

Working through it

The setup for the part. \$48 was convenient, not typical. Demand swings, and a price taker receives whatever number the market prints.

The three zones, defined by the two trip-wires.

- $P > 48$ (above min ATC): the firm operates at a **profit**.
- $20 \leq P < 48$: the firm operates at a **loss** — and does so deliberately, which is the least intuitive result in the part.
- $P < 20$ (below min AVC): the firm **shuts down**.

Why the middle zone exists at all. Because in the short run the fixed cost is owed whether the firm operates or not. So the comparison is not "am I making money?" but "does operating lose me less than not operating?" — and between 20 and 48 the answer is yes.

The plan. One concrete price per zone: \$71, then \$31, then \$15. Each is worked both numerically and geometrically, because the rectangle geometry is what an exam asks you to draw.

Scenario 1: $P = 71$ — profit

Scenario 1: $P = 71$ — profit

Marginal route: unit 7 costs $59 \leq 71$; unit 8 costs $85 > 71$. So $q^* = 7$.

Total route	$\pi = TR - TC = 71 \times 7 - 347 = 497 - 347 = +\$\mathbf{150}$
Per-unit route	$\pi = (P - ATC) \times q = (71 - 49.57) \times 7 = 21.43 \times 7 = +\$\mathbf{150}$

Two routes, same \$150 — and the per-unit route is exactly the **rectangle** we are about to draw.

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Slide 30

Working through it

The output decision. Run up the marginal cost column against \$71. Unit 7 costs \$59, which the price covers; unit 8 costs \$85, which it does not. So $q^* = 7$ — the higher price has drawn out one more unit than \$48 did.

Profit by the total route.

$$\pi = TR - TC = 71 \times 7 - 347 = 497 - 347 = +\$\mathbf{150}$$

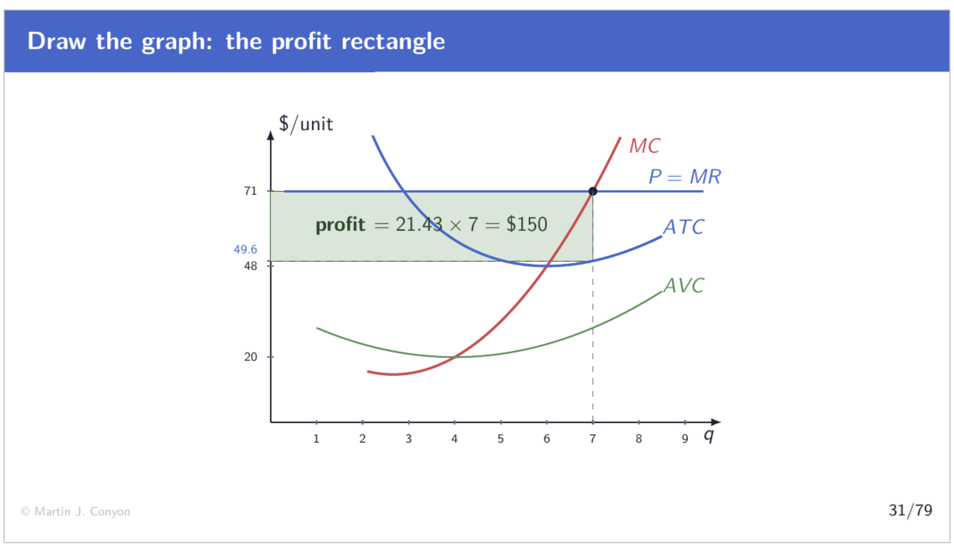
Profit by the per-unit route. ATC at $q = 7$ is $347/7 = 49.57$, so

$$\pi = (P - ATC) \times q = (71 - 49.57) \times 7 = 21.43 \times 7 = \$\mathbf{150}$$

Why the second route matters more than the first. It is *geometric*. $P - ATC$ is a vertical distance on the cost diagram and q is a horizontal one, so profit is literally the area of a rectangle — which is what the next frame draws.

One caution about ATC. It must be read at the *chosen* output, 7, not at the minimum. ATC at $q = 7$ is \$49.57, not \$48. Using the minimum is the commonest way to get the rectangle wrong.

Draw the graph: the profit rectangle



Slide 31

Working through it

How to construct the rectangle, in order.

1. Find the **dot**: where the price line meets marginal cost. That fixes $q^* = 7$.
2. Drop from the dot to the ATC **curve** at that same output. The vertical gap is profit per unit, \$21.43.
3. The rectangle runs from $q = 0$ to $q = 7$, with its top at the price line and its bottom at ATC.

Area $= 21.43 \times 7 = \$150$.

The two errors that cost marks. Reading ATC at its minimum instead of at the chosen output; and drawing the rectangle out to where ATC is lowest rather than to q^*. Both come from forgetting that the output is chosen by MC, and the rectangle is then measured *there*.

Why the picture is worth the trouble. Once the construction is automatic, the loss case is the identical construction with the price line below ATC, and the zero-profit case is the price line exactly touching ATC at its minimum — three scenarios, one drawing.

Scenario 2: $P = 31$ — a loss, taken on purpose

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Marginal route: unit 5 costs $25 \leq 31$; unit 6 costs $39 > 31$. So $q^* = 5$, and $\pi = 31 \times 5 - 249 = -\$94$. Why not quit?

	operate at $q = 5$	shut down
revenue	155	0
variable cost	105	0
fixed cost	144	144
result	**−\$94**	−\$144

−94 beats −144: revenue covers all the variable cost, with \$50 left toward the rent.

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Slide 32

Working through it

The output decision. Against \$31, unit 5 costs \$25 (covered) and unit 6 costs \$39 (not). So $q^* = 5$, and

$$\pi = 31 \times 5 - 249 = 155 - 249 = -\$94$$

A loss. The instinct is to stop, so the frame tests that instinct properly.

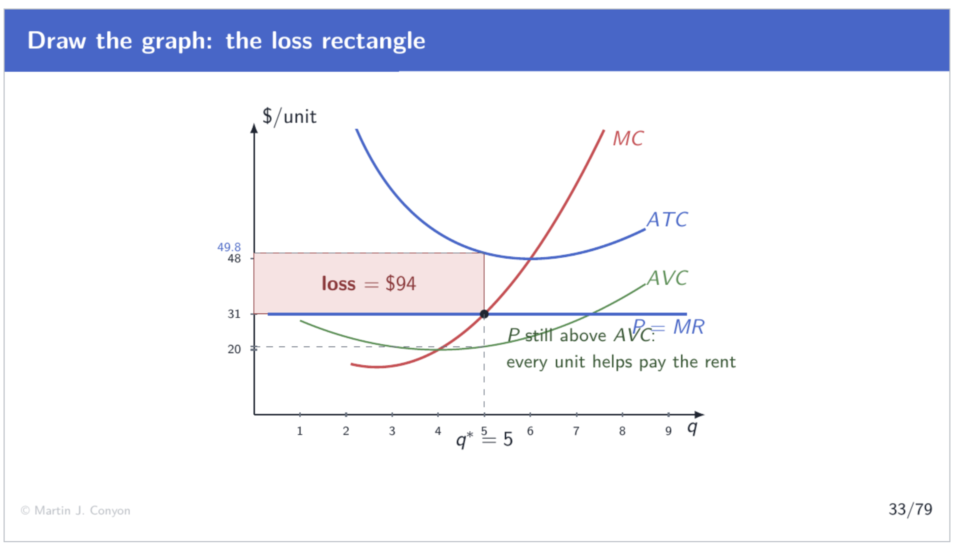
The comparison that actually matters.

- **Operate at** $q = 5$: revenue \$155, variable cost \$105, fixed cost \$144. Result: $155 - 105 - 144 = -\$94$.
- **Shut down**: revenue \$0, variable cost \$0, fixed cost \$144 still owed. Result: $-\$144$.

So operating is better by \$50. And the \$50 has a precise identity: revenue of \$155 covers all \$105 of variable cost and leaves \$50 over, which goes toward the rent. Fifty dollars of the fixed bill paid is fifty dollars less lost.

The general principle. Operating at a loss is correct whenever revenue covers variable cost, because every dollar above variable cost is a dollar the firm would otherwise have to find elsewhere. "Losing money" and "should stop" are different statements.

Draw the graph: the loss rectangle



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Working through it

The same construction, opposite sign. The dot is where the price line meets marginal cost, at $q = 5$. ATC at that output is $249/5 = \$49.80$, which is *above* the \$31 price, so the rectangle hangs between the price line and ATC: height \$18.80, width 5, area \$94 — the loss.

The other reading, and it is the one that decides the case. Look at where the price line sits relative to AVC. At $q = 5$, $AVC = \$21$, and the price is \$31 — so the price line is *above* the green curve.

What that gap represents. $(31-21)\times 5 = \$50$: exactly the contribution toward the fixed cost computed on the previous frame. The geometry and the arithmetic are the same statement.

The rule made visual. As long as the price line lies above AVC somewhere, there is a positive contribution and the firm operates. The moment the price line sinks below the entire AVC curve, no output produces any contribution at all — and that is Scenario 3.

Scenario 3: $P = 15$ — shut down

Scenario 3: $P = 15$ — shut down

The cheapest the firm can *ever* make connectors, per unit of variable cost, is $AVC = 20$ (at $q = 4$).

At $P = 15$, **every** possible output sells below its variable cost: each unit produced deepens the loss. Best choice: $q = 0$, lose the fixed bill (\$144), and no more.

Shutdown rule. In the short run, operate if $P \geq \min AVC$; if $P < \min AVC$, produce zero. Fixed cost plays no part in the decision.

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Working through it

Why this price is different in kind. The lowest AVC the technology can ever achieve is \$20, at $q = 4$. A price of \$15 is below that — so there is *no* output at which revenue covers even the variable cost of producing it.

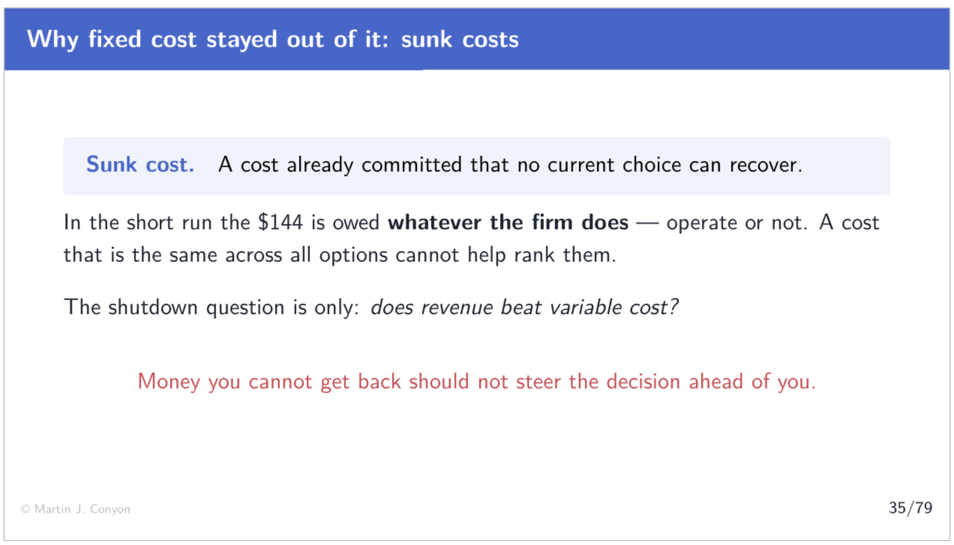
The consequence. Every connector made at \$15 loses money on its own materials and labor, before the rent is mentioned at all. Producing more deepens the loss rather than defraying it.

The best available choice. Produce zero, and lose the fixed bill: $-\$144$ and no worse.

The shutdown rule. In the short run, operate if $P \geq \min AVC$; produce zero if $P < \min AVC$. Fixed cost plays no part in the decision.

What is conspicuously absent from that reasoning. The \$144 itself. It was never compared with anything, because it is owed under both options and therefore cannot distinguish them. That omission is not sloppiness — it is the whole content of the next frame.

Why fixed cost stayed out of it: sunk costs



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Working through it

The definition. A **sunk cost** is a cost already committed that no current choice can recover.

Why sunk costs are irrelevant to decisions. A cost that is identical across all the options cannot make one option better than another. It cancels out of every comparison. In the short run the \$144 is owed whether the firm operates or not, so it cancels — which is exactly why the shutdown rule compares price with *variable* cost only.

The general form of the error. Letting unrecoverable past spending influence a forward-looking choice. Finishing a bad meal because it was expensive. Sitting through a dreadful film because the ticket is paid for. Continuing a failing project because of what has already been spent on it.

Why the pull is so strong. It feels like waste to walk away from money already spent. But the money is gone under *both* options — the only live question is which future is better, and the sunk cost is in neither future.

The firm's version, stated cleanly. The rent is gone either way. So only revenue against variable cost can decide whether to run the line.

5. Part 5: One Rule, Seen Twice

Collect the answers

Collect the answers

Every scenario asked the same question — *how much, at this price?* Line the answers up:

P	q supplied	from
15	0	below min AVC: shut
20	4 (or 0 — indifferent)	the knife edge
31	5	$P = MC$
48	6	$P = MC$
71	7	$P = MC$

A list of quantities offered at every price has a name: **a supply schedule**. We built the firm's supply curve without noticing.

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Working through it

What the scenarios actually produced. For each price the market printed, one best quantity:

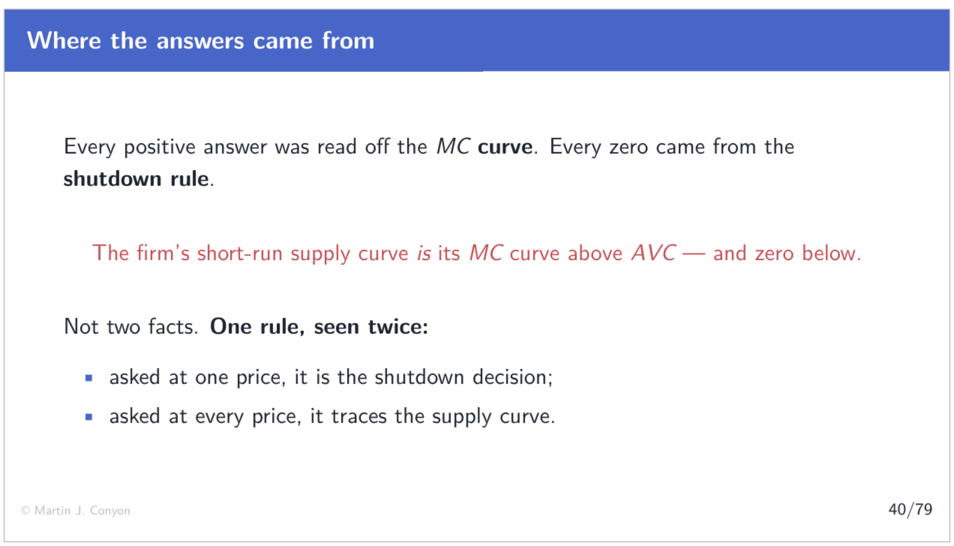
$$15 \to 0 \qquad 20 \to 4 \text{ (or } 0) \qquad 31 \to 5 \qquad 48 \to 6 \qquad 71 \to 7$$

What that list is. A menu of quantities offered at every price — which is the definition of a **supply schedule**.

The point of the frame. Nobody set out to derive a supply curve. Five ordinary business decisions were answered — how much to make at this price? — and the supply curve fell out of the answers as a by-product.

Why that matters conceptually. Supply has been used since the very outset as though it were a primitive assumption about seller behavior. It is not. It is profit maximization, tabulated across prices. The upward slope is not assumed; it is inherited from rising marginal cost, which was itself inherited from diminishing marginal product.

Where the answers came from



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Working through it

Sorting the answers by their source. Every *positive* quantity came from riding the price along the MC curve. Every *zero* came from the shutdown rule.

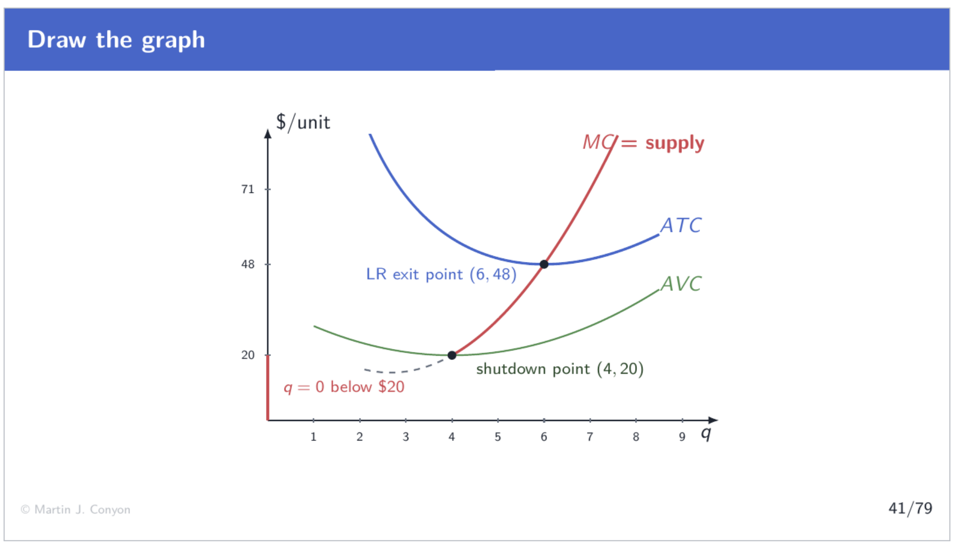
So the supply curve is. The MC curve above AVC, and zero below.

The claim being made, and it is the title of the part. These are not two facts to memorize. They are **one rule asked in two different ways**:

- Asked at *one* price, it answers a yes-or-no business question — the shutdown decision.
- Asked at *every* price at once, it traces a curve — supply.

Why students lose marks here. Learning "the shutdown rule" and "the supply curve" as separate items invites questions like "how do they fit together?" — which has no good answer because they were never apart. The shutdown rule does not sit alongside the supply curve; it *draws* the lower end of it.

Draw the graph



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Working through it

The supply curve, assembled from three pieces.

1. **Below \$20**: the curve runs up the vertical axis at $q = 0$. No price in this range produces any output at all.

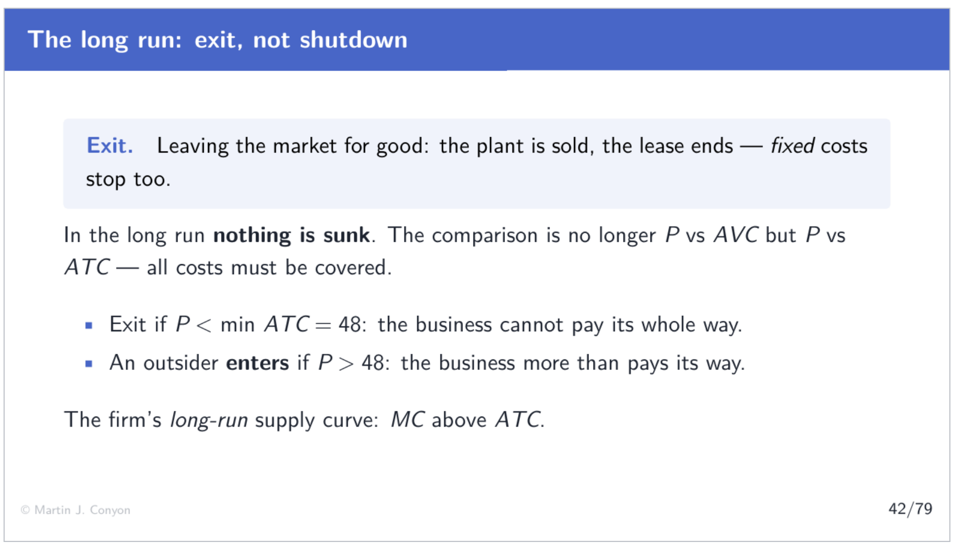
2. **At \$20**: it jumps horizontally out to the shutdown point $(4, 20)$ — the firm goes from producing nothing to producing four.
3. **Above \$20**: it *is* the marginal cost curve, climbing through $(5, 31)$, $(6, 48)$, $(7, 71)$.

Why the jump. At exactly \$20 the firm is indifferent between 0 and 4, so the supply correspondence contains both. A penny above and 4 strictly wins; a penny below and 0 strictly wins. The discontinuity is real, not a drawing artefact.

The dashed segment. The part of MC below AVC is genuine arithmetic — those marginal costs exist — but no price ever puts the firm there, so it is economically dead. Drawing it dashed records both facts.

The blue dot higher up. Where MC crosses the minimum of ATC at $(6, 48)$. That is not a short-run trip-wire at all; it is the long-run one, and the next frame explains why.

The long run: exit, not shutdown



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Working through it

Two different decisions, often confused.

- **Shutdown** is a short-run posture: produce nothing, keep paying the rent, wait for the price to recover. The plant still exists.

- **Exit** is final: the plant is sold, the lease ends, and the fixed cost stops with it.

Why the trip-wire moves. In the short run the \$144 is sunk, so it cannot enter the comparison and the test is P against AVC. In the long run *nothing* is sunk — every cost is avoidable by leaving — so the test becomes P against ATC.

- **Exit** if $P < \min\ ATC = 48$: the business cannot pay its whole way.
- An outsider **enters** if $P > 48$: the business more than pays its whole way.

The symmetry worth noticing. The same \$48 threshold governs both directions. Read from inside the market it is an exit warning; read from outside it is an invitation. Entry and exit are traffic through one door — and that traffic is the engine of Part 7.

The long-run supply curve of the firm. MC above ATC, since below \$48 the firm is not in the market at all.

The decision rules, one table

The decision rules, one table

Price	Short run	Long run
$P < 20$	shut down	exit
$20 \le P < 48$	operate at a loss	exit
$P = 48$	operate, $\pi = 0$	stay — and no one enters
$P > 48$	operate, $\pi > 0$	stay — and **entrants come**

Short run: P against **min** AVC (can revenue beat variable cost?).
Long run: P against **min** ATC (can revenue beat *all* cost?).

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Working through it

The whole decision map, and it is two comparisons.

- **Short run**: P against min $AVC = 20$. Can revenue beat *variable* cost?
- **Long run**: P against min $ATC = 48$. Can revenue beat *all* cost?

Reading the four rows.

- $P < 20$: shut down now, exit later. Both tests fail.
- $20 \leq P < 48$: *operate at a loss* now, exit later. The interesting row — the two runs give opposite instructions, and both are correct.
- $P = 48$: operate with zero profit; stay, and nobody enters.
- $P > 48$: operate at a profit; stay — **and entrants come**.

The row that generates the second half. The last one. A price above $48 is not merely good news for the incumbent; it is a signal visible to everybody outside the market, and free entry means outsiders can act on it. What their arrival does to the price is Part 7.

6. Part 6: From Firm to Market

Add the firms up

Add the firms up

Short run: the cast is fixed — **120 identical firms**, no one can enter or leave yet.
Market supply at each price is $120 \times q$:

P	each firm's q	market $Q^s = 120 \times q$
20	4	$120 \times 4 = 480$
31	5	$120 \times 5 = 600$
48	6	$120 \times 6 = 720$
71	7	$120 \times 7 = 840$

The market supply curve is the firm's, stretched 120 times wider.

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Working through it

The short-run assumption. The cast is fixed at **120 identical firms**. Nobody can enter or leave yet, because building or dismantling a plant takes time.

Market supply is then plain addition. At each price, take what one firm supplies and multiply by 120:

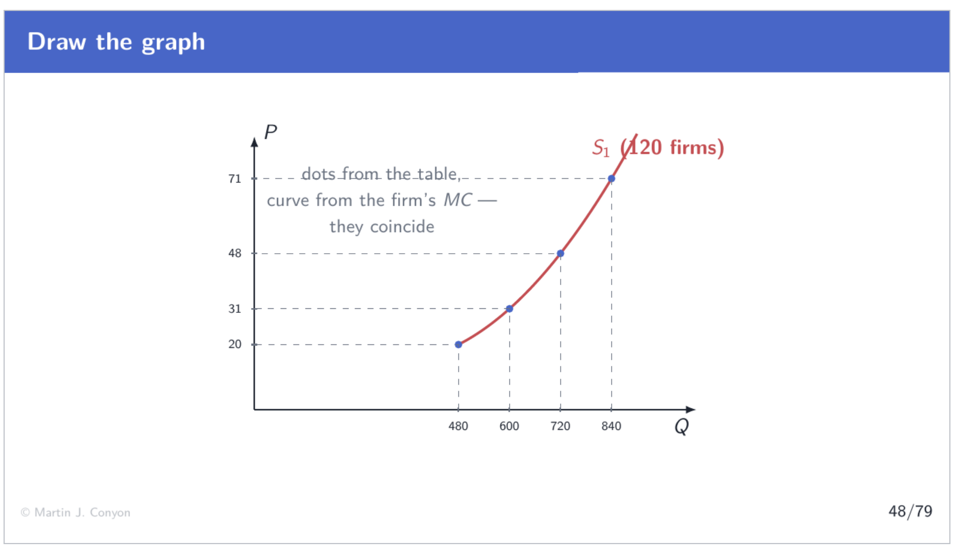
$$P = 48: \ 120 \times 6 = 720 \qquad P = 71: \ 120 \times 7 = 840$$

Why horizontal addition is the right operation. Every firm faces the same price, so the question at each price is how much they collectively offer — which is the sum of their individual quantities. Adding prices at a fixed quantity would answer nothing.

What the market curve inherits. Everything. It slopes up because each firm's MC slopes up. It begins at 480 because below $20 every firm shuts down at once — all 120 are identical, so they all hit the trip-wire together.

The one thing to hold on to. The market supply curve is not a separate object with its own properties. It is one firm's marginal cost curve with the quantity axis stretched by the head count.

Draw the graph



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Working through it

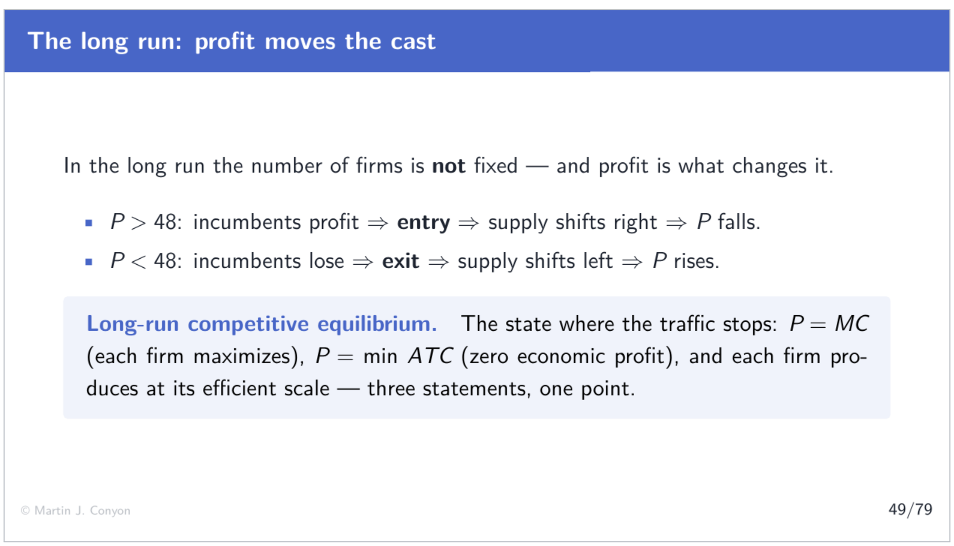
What is drawn. Market supply with 120 firms: dots from the addition table, curve from the firm's marginal cost function with quantity multiplied by 120.

Why dots and curve agree exactly. They are the same object computed two ways. The dots are $120 \times q$ at four specific prices; the curve is the same multiplication performed at every price. Any disagreement would be an arithmetic error, not a modeling choice.

Reading the shape. It starts at $Q = 480$ at a price of \$20 — the market's own shutdown point, where all 120 firms simultaneously become willing to produce 4 each. Above that it rises, steeply at first and then more so, because it inherits the curvature of MC.

What has just been earned. The market supply curve was drawn on faith at the very outset — assumed to slope upward because that seemed reasonable. It has now been built from its parts, and the upward slope is a derived consequence of diminishing marginal product. Nothing about supply remains assumed.

The long run: profit moves the cast



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Working through it

The change. In the long run the number of firms is not fixed, and *profit* is what changes it.

The two directions.

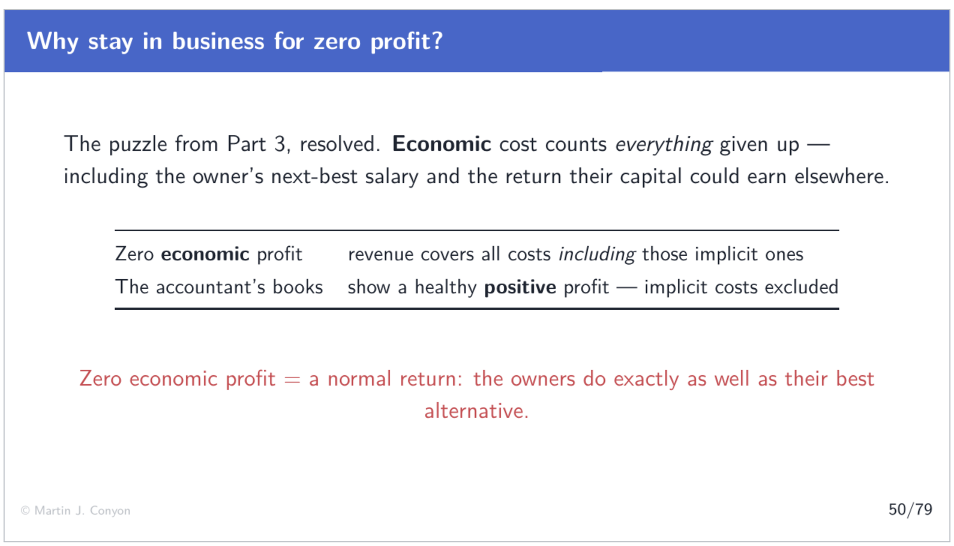
- $P > 48$: incumbents profit $\Rightarrow$ outsiders **enter** $\Rightarrow$ supply shifts right $\Rightarrow$ price **falls**.
- $P < 48$: incumbents lose $\Rightarrow$ firms **exit** $\Rightarrow$ supply shifts left $\Rightarrow$ price **rises**.

Both arrows point at \$48, which makes it a stable resting point rather than a coincidence.

The definition of long-run competitive equilibrium. Three statements: $P = MC$ (each firm maximizes profit), $P = \min ATC$ (zero economic profit, so no entry or exit), and each firm produces at its *efficient scale.*

Why the three are one point, not three conditions. On the cost diagram, MC crosses ATC exactly at the minimum of ATC. So a price equal to min ATC automatically meets MC there too, and that quantity is by definition efficient scale. One point on one diagram satisfies all three.

Why stay in business for zero profit?



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Working through it

The objection, and it deserves a straight answer. If competition grinds profit to zero, why would anybody run a business at all?

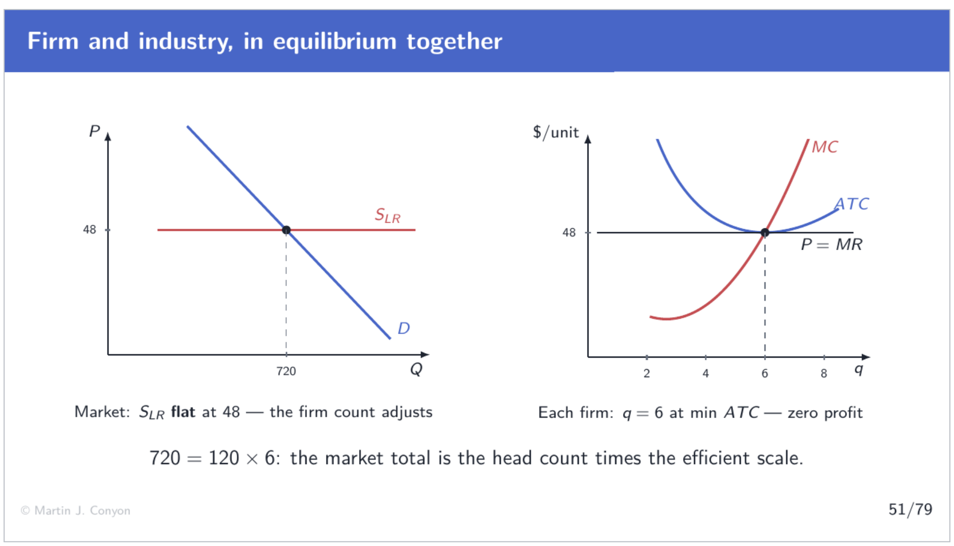
The resolution is in which *zero.* Economic cost includes every opportunity cost — explicitly, the owner's next-best salary and the return her capital could earn elsewhere. So:

- **Zero economic profit** means revenue covers all costs *including* those implicit ones. The owner is being paid her full market worth and her capital is earning its full market return, inside this firm.
- **The accountant's books** show a healthy *positive* profit, because implicit costs never appear on them.

So zero economic profit is a normal return. The owner does exactly as well here as in her best alternative — which is precisely why she neither leaves nor is joined by imitators.

Why the concept is worth the trouble. Only economic profit predicts entry and exit correctly. Active Learning 6 supplies a firm with an $80,000 accounting profit that should nevertheless close, and the two measures give opposite verdicts.

Firm and industry, in equilibrium together



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Working through it

Two panels, and Part 7 is nothing but this pair in motion — so learn to read them together now.

Left panel: the industry. The long-run supply curve is **horizontal at $48**. That deserves an explanation: the market can deliver any total quantity at $48, not by squeezing more out of each firm, but by changing *how many firms* there are. Each firm always produces 6; the total is however many firms times 6. Demand crosses this flat line at $Q = 720$.

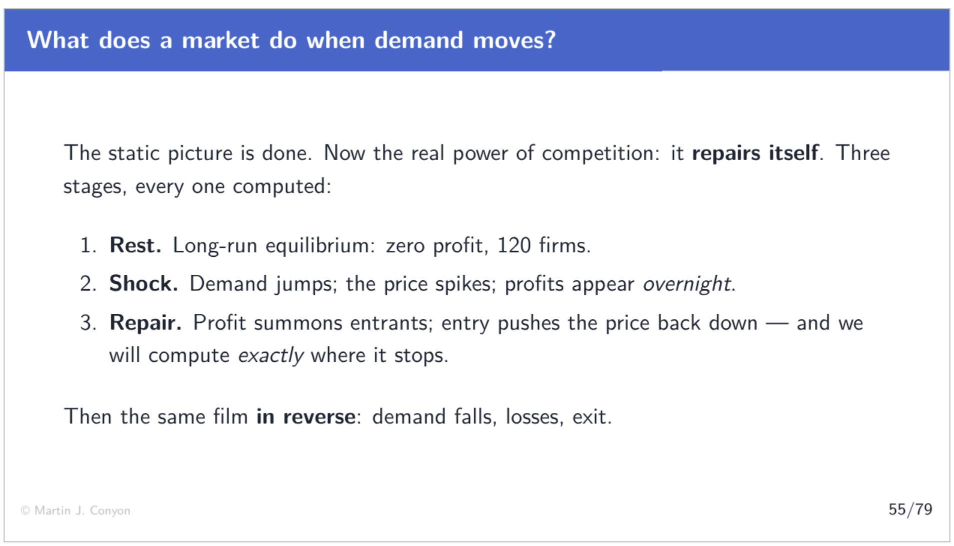
Right panel: one firm. It takes the $48 price as a flat line, meets its marginal cost at $q = 6$, and sits exactly at the bottom of its ATC curve — so the profit rectangle has zero height.

How the panels lock together. $720 = 120 \times 6$. The market total is the head count times the efficient scale, so every number on the left is built from the numbers on the right.

What to watch for in Part 7. A demand shock moves the left panel first; the right panel then shows what the new price does inside a firm; and the firm's profit or loss determines what happens to the head count, which moves the left panel again.

7. Part 7: The Dynamics of Competition

What does a market do when demand moves?



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Working through it

What changes now. Everything so far has been a still photograph of a market at rest. This part sets it in motion, and it is the most important material in the topic.

The three stages.

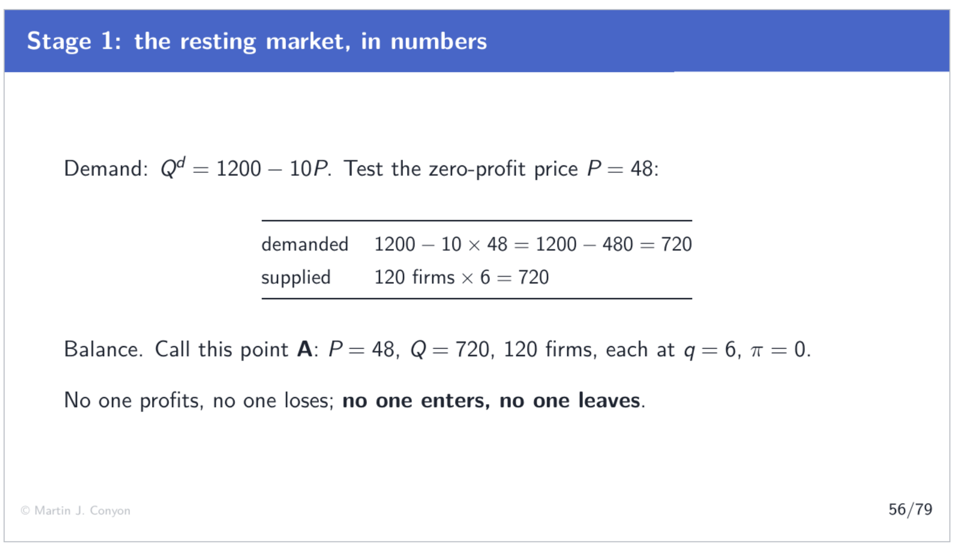
1. **Rest.** Long-run equilibrium: \$48, zero profit, 120 firms.
2. **Shock.** Demand jumps. With the cast frozen, the price spikes and profits appear overnight.
3. **Repair.** Profit summons entrants; entry pushes the price back down — and the stopping point is computed exactly, not asserted.

Then the same film in reverse: demand falls, losses appear, exit repairs it.

What makes this treatment different from a verbal one. At every stage the price, each firm's output, each firm's profit and the number of firms are all computed. There are no arrows of faith. In particular, "entry continues until profit is zero" is refined into an exact whole-firm calculation, because a third of a firm cannot enter.

Why the reverse film gets equal time. Exam answers regularly get the falling-demand case wrong by assuming firms shut down at the first loss. They do not, and Part 4 already showed why.

Stage 1: the resting market, in numbers



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Working through it

Setting the scene with arithmetic rather than adjectives. Demand is $Q^d = 1200 - 10P$.

Testing \$48, the only candidate for a long-run price.

$$\text{demanded: } 1200 - 10(48) = 720 \qquad \text{supplied: } 120 \times 6 = 720$$

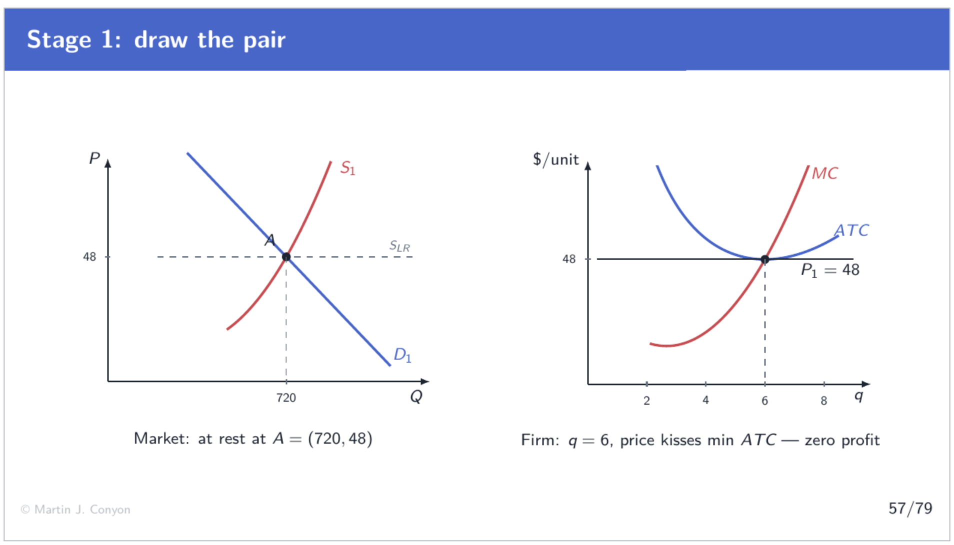
It balances.

Why \$48 is the only candidate. At any other price the firms are making either a profit or a loss, so entry or exit would already be under way and the market would not be at rest. Long-run equilibrium requires zero economic profit, which requires $P = \min ATC$.

Point A, fully specified. $P = 48$, $Q = 720$, 120 firms, each producing $q = 6$, each earning $\pi = 0$.

Why nothing moves. No profit to attract entrants; no loss to drive incumbents out; each firm already at its best output for this price. The market is at rest in every sense, and it stays there until something outside it changes.

Stage 1: draw the pair



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Working through it

The two panels at rest.

Left, the market. Short-run supply from 120 frozen firms crosses demand at $(720, 48)$.

Right, one firm. The price line sits at \$48, meets MC at $q = 6$, and touches ATC exactly at its minimum — so there is no gap between price and average cost, and therefore no profit rectangle to draw.

The link between the panels. The quantity on the left is the quantity on the right multiplied by the head count: $720 = 120 \times 6$. That relationship holds at every stage of this part, and checking it is a useful way to catch an error.

What to watch as the film runs. Three things move, in a fixed order. First the *price*, set on the left by demand meeting a frozen supply. Second the *profit rectangle*, which opens or closes on the right as the price line moves relative to ATC. Third the *number of firms*, which responds to that rectangle and shifts the left panel's supply curve. Never the other way round.

Stage 2: demand jumps

Stage 2: demand jumps

A new technology needs connectors everywhere. Demand shifts to $Q^d = 1550 - 10P$.

At the old price $P = 48$: demanded $= 1550 - 480 = 1070$, supplied $= 720$. Shortage of 350 — the price must rise.

Find the new short-run balance (cast still frozen at 120):

try P	demanded	supplied (120 firms)
48	1070	$120 \times 6 = 720$ — shortage
71	$1550 - 710 =$ **840**	$120 \times 7 =$ **840** — **balance**

Point **B**: $P = 71$, $Q = 840$.

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Working through it

The shock. A new technology needs connectors everywhere, and demand shifts to $Q^d = 1550 - 10P$ — at any price, buyers now want 350 more than before.

Why the price must rise. At the old price of \$48, demand is $1550 - 480 = 1070$ while supply is still 720. A shortage of 350, and a shortage bids the price up.

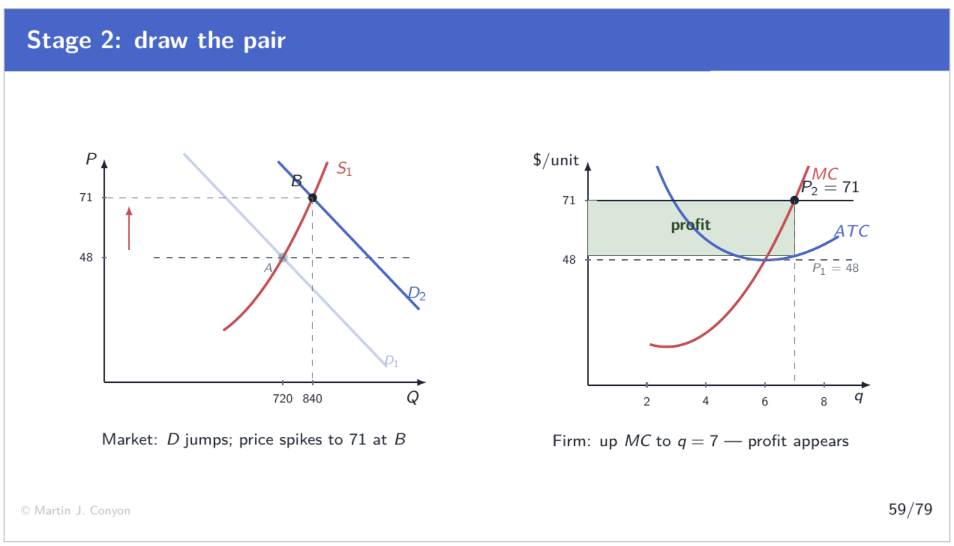
Finding the new short-run balance. The cast is frozen at 120 firms — no plant can be built overnight — so the price must rise along their *existing* supply schedule until demand and supply meet:

$$P = 71: \quad \text{demanded } 1550 - 710 = 840 \qquad \text{supplied } 120 \times 7 = 840$$

Note what each firm did. At \$71 the seventh unit becomes worth making (its MC is \$59), so every firm expands from 6 to 7. The market's extra 120 units come from existing firms working harder, since no others exist yet.

Point B. $P = 71$, $Q = 840$, still 120 firms. And \$71 is exactly the profit scenario rehearsed in Part 4.

Stage 2: draw the pair



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Working through it

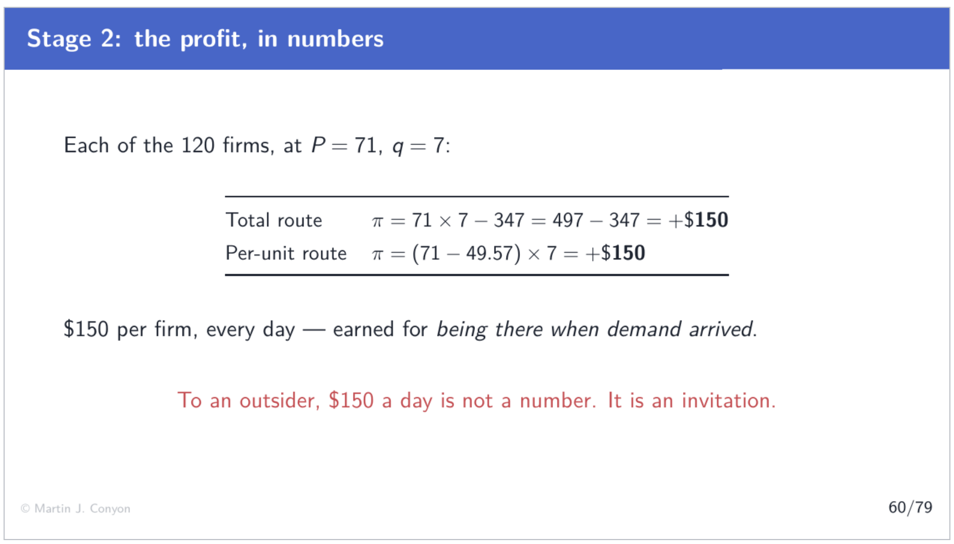
Left panel. The demand curve has shifted right, and because supply is frozen the equilibrium climbs *along* the existing supply curve to $(840, 71)$. The steepness of the short-run supply curve is why the price rises so far — 120 firms can only stretch so much.

Right panel. The firm's flat price line has risen to \$71. It now meets MC further out, at $q = 7$. And crucially, ATC at $q = 7$ is \$49.57, well *below* the price — so a profit rectangle opens between them.

The order of causation, visible in the pictures. The market panel moved the line; the firm panel shows what the line does once it arrives. The firm did not choose a higher price; it received one and responded by producing more.

What has not yet happened. Nothing has entered. The head count is unchanged at 120, which is why this is a *short-run* equilibrium. The profit rectangle on the right is the thing that will change that.

Stage 2: the profit, in numbers



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Working through it

Each firm's profit at $P = 71$, $q = 7$, by both routes.

$$\text{totals: } 71 \times 7 - 347 = 497 - 347 = +\$\mathbf{150}$$

$$\text{per unit: } (71 - 49.57) \times 7 = +\$\mathbf{150}$$

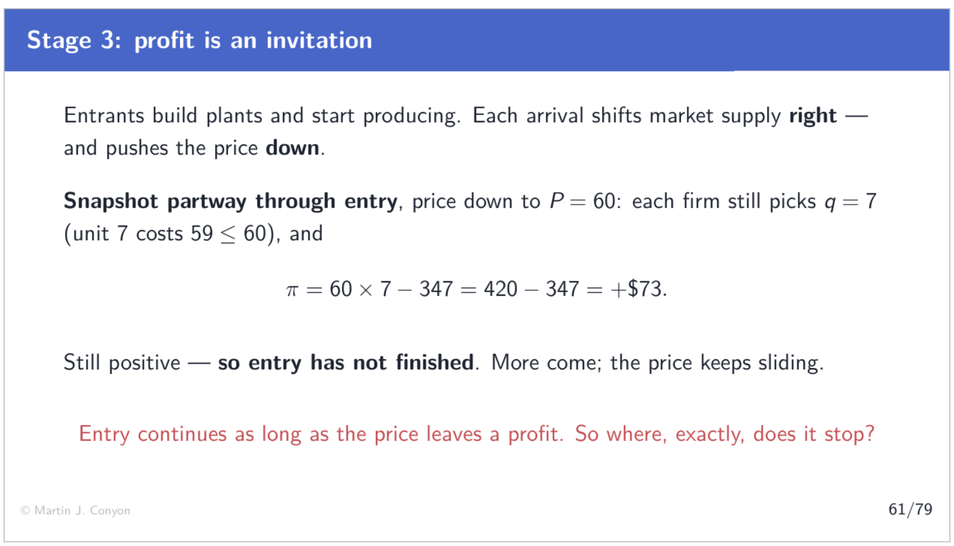
What the incumbents did to earn it. Nothing, in a sense — they were already there when demand arrived. The $150 a day is a return to being in the market at the moment it expanded.

The two things the number is doing at once, and the second is what matters.

- **Inside** the market it is a reward to incumbents.
- **Outside** the market it is *information*. It announces to anyone with capital and a blueprint that connector plants currently pay better than the next-best use of the same money.

Why the second function is the operative one. Free entry — condition 3 of the definition, back in Part 1 — means outsiders can act on that announcement. Profit is not merely a reward in a competitive market; it is a recruiting signal, and Stage 3 is what happens when it is answered.

Stage 3: profit is an invitation



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Working through it

The mechanism. Outsiders observe $150 a day, build plants, and start producing. Each arrival adds its own output to the market, so *market supply shifts right.*

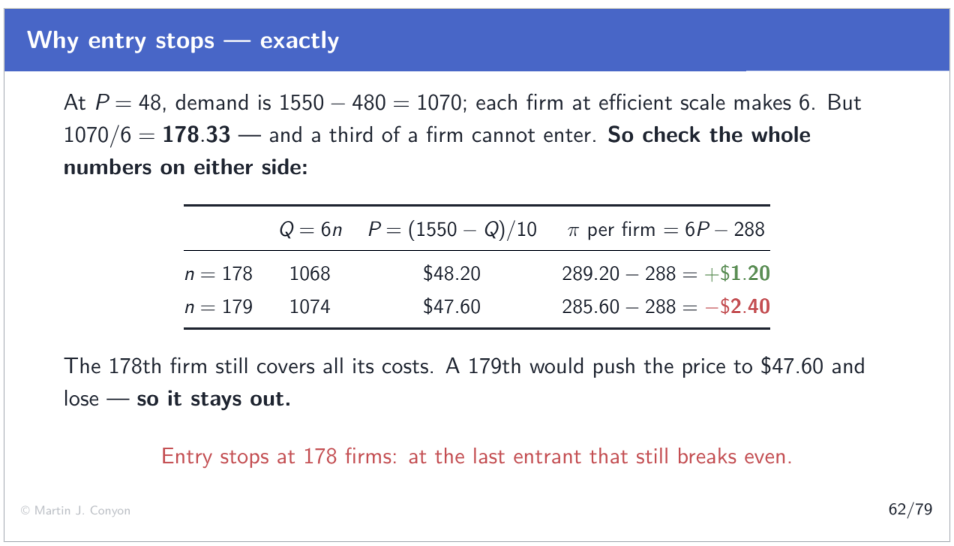
What a rightward supply shift does. Against an unchanged demand curve, it pushes the equilibrium price *down*.

The self-limiting property, which is the elegant part. Every entrant erodes the very profit that attracted it. Entry is not a stampede that overshoots and then corrects; each additional entrant makes the market slightly less attractive to the next one.

The question that remains, and it is a genuine one. Where does this stop? The loose answer — "when profit reaches zero" — is nearly right and not quite, because firms come in whole units. A price that yields exactly zero profit would require 178.33 firms, and a third of a firm cannot enter.

So the next frame does the arithmetic properly, testing the whole numbers on either side and showing that the stopping point is decided by a potential entrant's forecast of what its *own* arrival would do to the price.

Why entry stops — exactly



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Working through it

The starting calculation. If the price returned all the way to \$48, demand would be $1550 - 480 = 1070$, and firms at efficient scale make 6 each: $1070/6 = 178.33$ firms. Not a whole number, so the true stopping point must be tested.

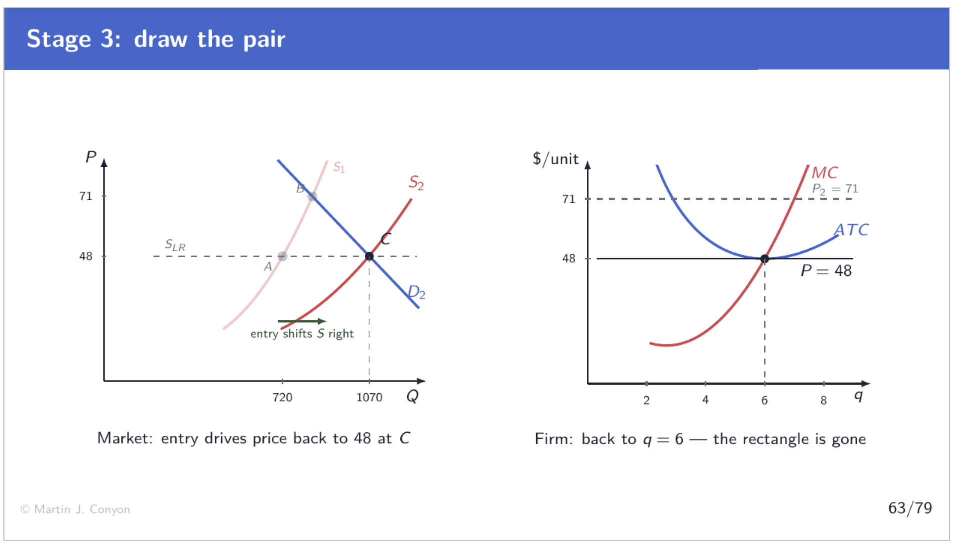
Testing 178 firms. $Q = 178 \times 6 = 1068$, so $P = (1550 - 1068)/10 = \$48.20$, and each firm earns $6 \times 48.20 - 288 = +\$\mathbf{1.20}$. Thin, but positive — these firms are covering all their costs.

Testing 179 firms. $Q = 1074$, so $P = (1550 - 1074)/10 = \$47.60$, and each firm earns $285.60 - 288 = -\$\mathbf{2.40}$. Everybody loses, including the newcomer.

Why the 179th firm stays out. Because it evaluates the market *as its own entry would leave it*, not as it currently is. Looking at \$48.20 and thinking "profitable, I'll join" would be a mistake: its arrival is precisely what destroys the profit. Foreseeing that, it declines.

So entry stops at 178 firms — at the last entrant that still breaks even. That is why the process converges rather than overshooting, and it is why the answer is a specific integer rather than a hand-wave.

Stage 3: draw the pair



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Working through it

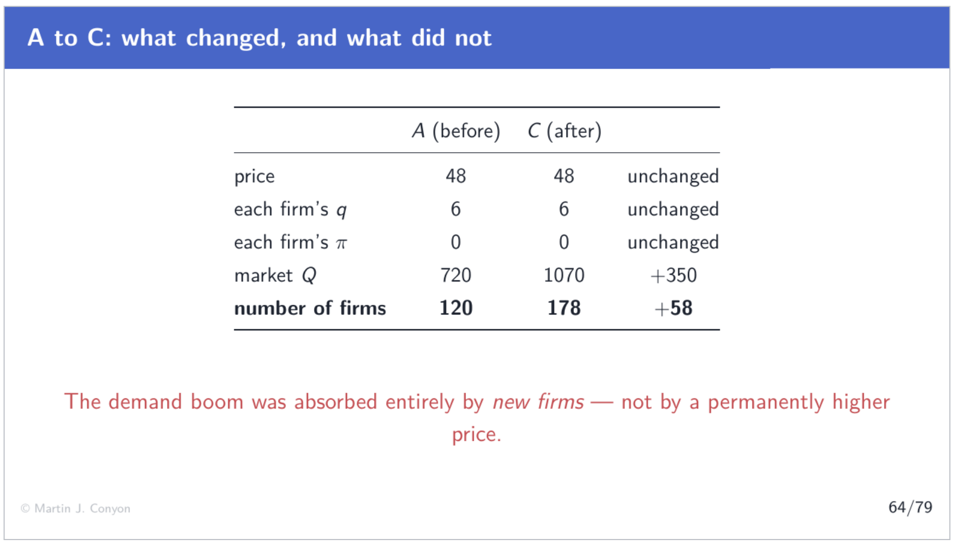
Left panel. Supply has swung right as 58 new plants arrived, and the new supply curve crosses the higher demand curve at a price back down near \$48. Note that the demand curve has *not* moved back — the boom is permanent. What changed is how much the industry can produce at each price.

Right panel. The firm's price line has sunk back to where it started, so its output returns to $q = 6$ and the profit rectangle closes to nothing. The incumbent ends exactly where it began.

The sequence, complete. Demand up → price up (frozen cast) → profit → entry → supply right → price back down → profit gone → entry stops. Each step causes the next; none is assumed.

What the incumbents got out of it. A temporary windfall of $150 a day for as long as the entry took. That is not nothing — it is the reward for being there when demand arrived — but it is temporary by construction.

A to C: what changed, and what did not


A to C: what changed, and what did not

	A (before)	*C* (after)	
price	48	48	unchanged
each firm's q	6	6	unchanged
each firm's π	0	0	unchanged
market Q	720	1070	+350
number of firms	**120**	**178**	+**58**

The demand boom was absorbed entirely by *new firms* — not by a permanently higher price.

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Working through it

The comparison, and the pattern in it is the point.

- **Price**: 48 → 48. Unchanged.
- **Each firm's output**: 6 → 6. Unchanged.
- **Each firm's profit**: 0 → 0. Unchanged.
- **Market quantity**: 720 → 1070. Up 350.

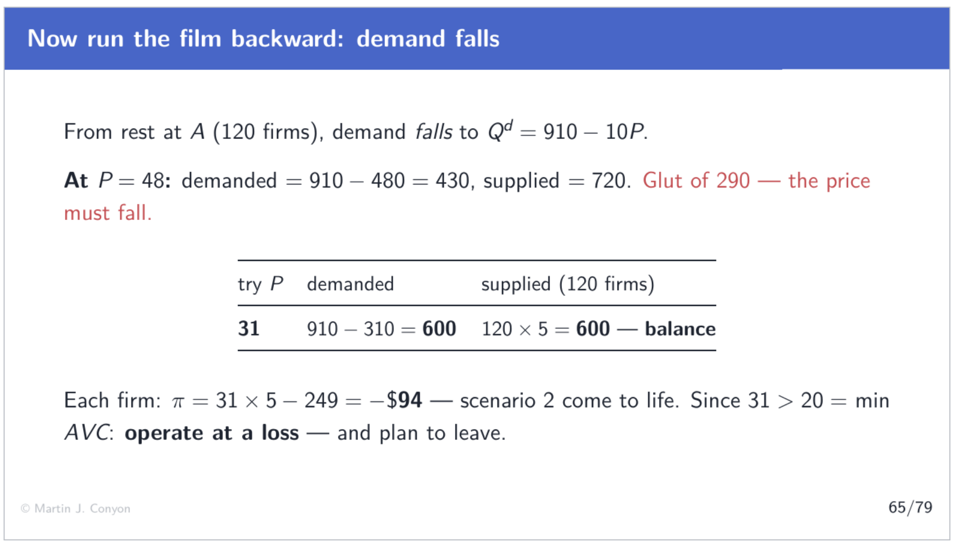
- **Number of firms**: 120 → 178. Up 58.

Reading it. Everything at the level of the individual firm is restored, as if the boom had never happened. The entire adjustment appears in the market total and the head count.

The slogan, and it is worth memorizing in this form. In the short run, demand shocks move the **price**. In the long run, they move the **number of firms**.

What the price spike was for. Not a failure of the market and not profiteering. It was the recruiting signal that built 58 new plants — and once they were built, it had done its job and disappeared. A market that could not raise its price temporarily would have no mechanism for summoning the capacity the extra demand required.

Now run the film backward: demand falls



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Working through it

The reverse shock. From rest at A, demand falls to $Q^d = 910 - 10P$.

Why the price falls. At \$48 demand is $910 - 480 = 430$ while 120 frozen firms still supply 720 — a glut of 290. A glut pushes the price down.

The new short-run balance. Sliding down the frozen supply schedule:

$$P = 31: \quad \text{demanded } 910 - 310 = 600 \qquad \text{supplied } 120 \times 5 = 600$$

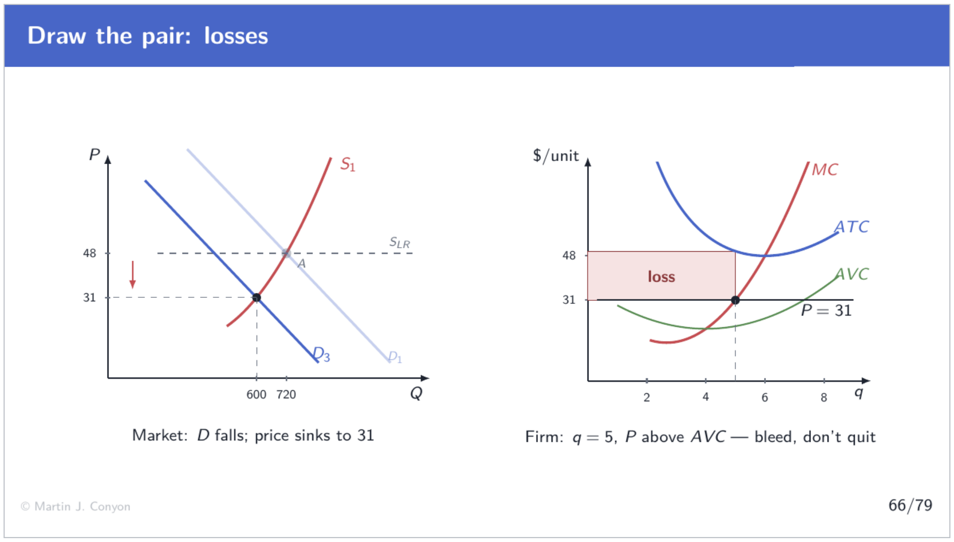
What each firm now faces. \$31 is exactly Scenario 2 from Part 4. Each firm produces 5 and loses

$$31 \times 5 - 249 = -\$94 \text{ a day}$$

The step that exam answers most often get wrong. The firms do *not* shut down. \$31 is comfortably above min $AVC = 20$, so every unit still contributes toward the fixed bill, and operating at $-\$94$ beats shutting at $-\$144$.

So how does the loss act? Slowly, and through a different channel: not by switching plants off, but by ushering firms out of the industry as leases expire.

Draw the pair: losses



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Working through it

Left panel. Demand has shifted left, and the equilibrium slides *down* the frozen supply curve to $(600, 31)$. The mirror image of the boom.

Right panel. The price line has dropped to \$31, meeting MC at $q = 5$. ATC at that output is \$49.80, above the price, so the rectangle between them is a *loss* of \$94.

The second reading, and it settles the operate-or-shut question. Look at where the price line sits against AVC. At $q = 5$, $AVC = \$21$ and the price is \$31, so the price line remains *above* the green curve. That gap — \$10 a unit on 5 units, \$50 — is the contribution toward the rent, and it is the geometric reason the plant keeps running.

The two rectangles to keep distinct. The loss rectangle (between P and ATC) says the firm is losing money. The contribution gap (between P and AVC) says it should nevertheless keep operating. Both are true at once, and each answers a different question.

Exit — and where it stops

Exit — and where it stops

Losses push firms out; each exit shifts supply **left**; the price **recovers**. At $P = 48$, demand is $910 - 480 = 430$, and $430/6 = \mathbf{71.67}$. **Whole firms again — check both sides:**

	$Q = 6n$	$P = \frac{910 - Q}{10}$	π per firm $= 6P - 288$
$n = 72$	432	\$47.80	$286.80 - 288 = \mathbf{-\$1.20}$ — still losing: *one more leaves*
$n = 71$	426	\$48.40	$290.40 - 288 = \mathbf{+\$2.40}$ — covered: *exit stops*

Same machine, opposite gear: losses expel firms until the survivors break even.

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Working through it

The mechanism in reverse. Losses push firms out as leases expire; each exit shifts market supply *left*; the price *recovers*.

The whole-firm calculation, exactly as for entry. If the price is to climb back to \$48, demand supports $910 - 480 = 430$ units, which is $430/6 = 71.67$ firms' worth. Test the integers on either side.

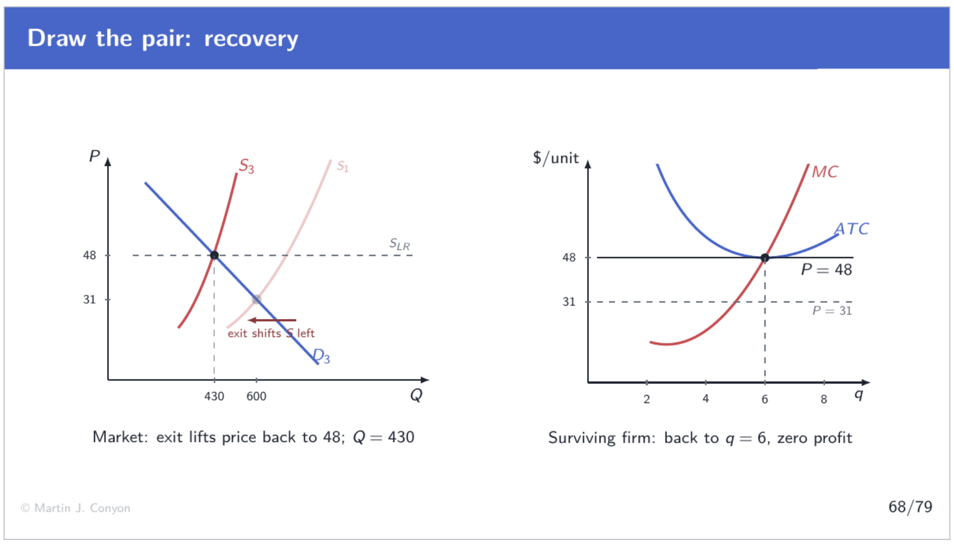
Testing 72 firms. $Q = 432$, $P = (910 - 432)/10 = \$47.80$, and each firm earns $286.80 - 288 = -\$\mathbf{1.20}$. Still losing — so another lease lapses and another firm goes.

Testing 71 firms. $Q = 426$, $P = (910-426)/10 = \$48.40$, and each firm earns $290.40-288 = +\$\mathbf{2.40}$. Above water — so the leaving stops.

The result. 49 plants have closed, and the 71 survivors are whole again at $q = 6$ and zero profit.

The symmetry with entry, which is exact. Both processes run until the *marginal* firm — the one deciding whether to enter, or the one deciding whether to stay — can just cover all of its costs. Same rule, opposite gear.

Draw the pair: recovery



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Working through it

Left panel. Supply has swung *left* as 49 plants closed, and it now crosses the lower demand curve back up at \$48. The industry is smaller; the price is where it always was.

Right panel. The survivor's price line has climbed back to \$48, its output returns to $q = 6$, and the loss rectangle closes to nothing. A firm that survived the contraction ends exactly as it began.

What absorbed the fall in demand. Not a permanently lower price — the price came all the way back. Forty-nine fewer plants.

The complete symmetry with the boom, worth stating explicitly. A demand increase was absorbed by 58 *more* firms at the same price. A demand decrease was absorbed by 49 *fewer* firms at the same price. In both cases the individual firm ends untouched, and the industry's size does all the adjusting.

The human cost the diagram hides. Forty-nine businesses closed, with whatever that meant for their owners and employees. The model says the adjustment is efficient; it does not say it is painless, and the two claims should not be confused.

Three results to keep

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Working through it

Three sentences, and every examinable question about competitive dynamics is one of them in costume.

1. Short run: the price moves. With the cast frozen, a demand shift changes the price, and the price changes profits — +\$150 in the boom, −\$94 in the slump.

2. Long run: profit returns to zero. Entry and exit grind it away, and the traffic stops precisely where the marginal firm breaks even — 178 firms in one direction, 71 in the other. That stopping point is computable, not approximate.

3. The lasting adjustment is in the number of firms. 120 → 178, or 120 → 71. The price returns to min $ATC = 48$ every time, like water finding its level.

The summary claim. Competition converts temporary profit into permanent capacity, and temporary loss into permanent downsizing. The price signal is the instrument; the change in industry size is the result.

8. Part 8: Beyond the Baseline

When the flat line tilts (1): scarce inputs

When the flat line tilts (1): scarce inputs

The horizontal S_{LR} leaned on a quiet assumption: **input prices hold still as the industry grows**.

Suppose connector plants need a specialized alloy in limited supply. Entry during a boom **bids up the alloy's price** — so *every* firm's cost curves shift up, and min ATC rises with the size of the industry.

Then a bigger market needs a *permanently* higher price to cover the higher cost:

the long-run supply curve slopes upward — though still flatter than short-run supply.

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Working through it

The assumption being examined. The horizontal long-run supply curve assumed that **input prices hold still as the industry grows**. That was never stated, and it is not always true.

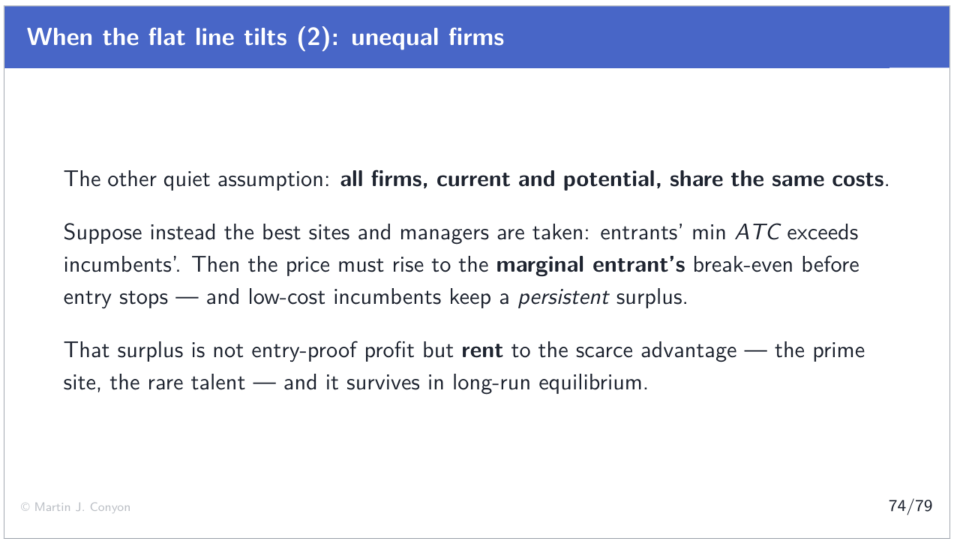
What happens if it fails. Suppose connector plants require a specialized alloy in limited supply. A wave of entry bids up the alloy's price. Every firm's cost curves shift *up*, so min ATC — and therefore the zero-profit price — is higher in a large industry than in a small one.

The consequence for the diagram. A bigger market needs a *permanently* higher price to cover the higher costs. The long-run supply curve slopes **upward**.

But still flatter than short-run supply. Entry continues to add capacity that a frozen cast cannot, so quantity still responds more to price in the long run than in the short run. Upward-sloping is not the same as steep.

Where you would see this. Industries leaning on something genuinely scarce — a mineral, a particular skill, land in one region. The flat line is the special case where the industry is small relative to the markets for its own inputs.

When the flat line tilts (2): unequal firms



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Working through it

The second assumption. That all firms, existing and potential, share the *same* costs. Convenient, and rarely true.

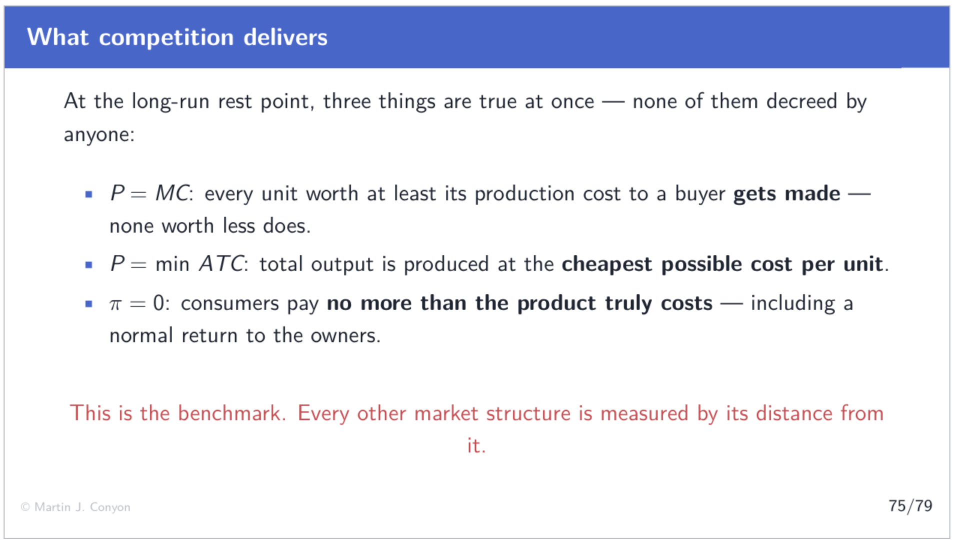
What happens if it fails. Suppose the best sites, the easiest customers and the most capable managers are already taken, so entrants face a higher min ATC than incumbents. Entry then continues until the price covers the **marginal entrant's** costs — the best of the firms still outside. Every firm cheaper than that keeps a surplus, permanently.

What to call that surplus, because the name matters. Not profit that entry somehow failed to erase. It is **rent** — the return to a scarce advantage that entrants cannot replicate.

Why the zero-profit logic survives. Charge the incumbent for its prime site at what that site is now worth, and its economic profit is zero after all. The advantage has a market value, and once that value is counted as a cost, the business itself earns a normal return like everyone else. Sell the farm and the buyer pays exactly for that stream.

The general lesson. Persistent surpluses in competitive industries are almost always rents to something scarce, and the analytical move is to ask what the scarce thing is and what it is worth.

What competition delivers



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Working through it

Three properties, all holding at once at the long-run rest point, and none of them decreed by anybody.

$P = MC$. Every unit worth at least its production cost to a buyer gets made, and no unit worth less does. The quantity is exactly right — this is the efficiency result from the surplus analyzis, arrived at from the firm's side.

$P =$ min ATC. Total output is produced at the lowest achievable cost per unit, because every firm sits at its efficient scale. Not merely the right quantity, but produced as cheaply as the technology allows.

$\pi = 0$. Buyers pay no more than the product truly costs, including a normal return to owners. No margin is extracted beyond what the resources are worth elsewhere.

Why this is called a benchmark rather than a description. Very few real markets satisfy all three conditions. The claim is not that markets look like this, but that this is the *standard* against which every other structure is measured — and the distance from it can be computed in dollars, as the monopoly analyzis does.

The other pole: one seller

The other pole: one seller

A **monopoly** is a market served by a single firm, protected from entry. It faces the *whole* downward-sloping market demand — to sell more it must **cut the price**, so its $MR < P$.

	Competition	Monopoly
sellers	many	one
demand facing firm	horizontal at P	the market's, sloping down
revenue logic	$MR = P$	$MR < P$
output rule	$P = MC$	$MR = MC$, so $P > MC$
long-run π	zero (entry)	can persist (entry blocked)

When we come to monopoly, the right-hand column is the whole story.

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Working through it

A first look at monopoly, using only tools already built. A monopoly is a market served by a single firm, protected from entry.

The one structural difference, from which everything else follows. It faces the *whole* market demand curve, which slopes down — not a flat line at a given price.

The chain of consequences.

- To sell one more unit it must **cut the price**, and the cut applies to every unit. So $MR < P$.
- Applying the universal rule $MR = MC$ therefore leaves $P > MC$: the price exceeds the cost of the last unit made.

- Blocked entry means profit is not competed away, so it can persist indefinitely.

Reading the comparison table. Every line of the monopoly column is the competitive column with *one* assumption removed — the assumption that the firm is too small to affect the price.

What $P > MC$ costs. If the price exceeds marginal cost, there are units buyers value above their production cost that do not get made. Those missing trades are a deadweight loss, and when we come to monopoly it is measured against exactly the benchmark established here.